Time-Series Analysis

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Topics

- 1. Introduction to Time-Series Analysis

- 2. Deep Learning Models for Time-Series Analysis

- 3. Case Studies

Section 1

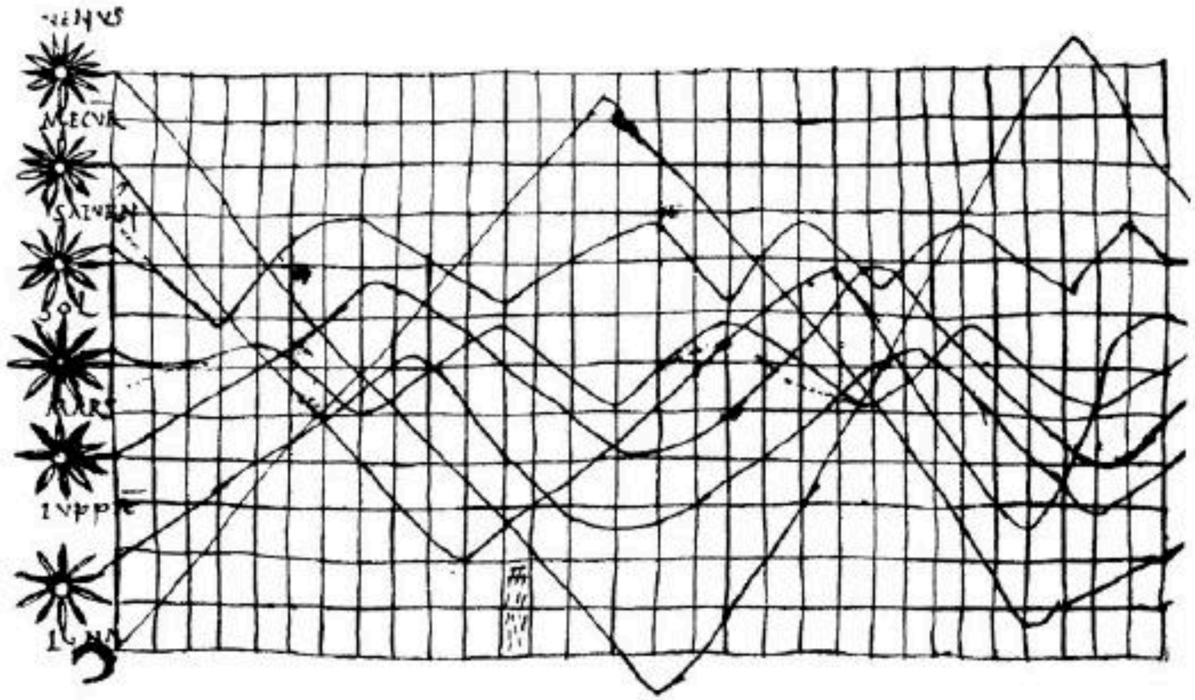
Introduction to Time-Series Analysis

시계열 분석

- **시계열 분석: 여러 시점에 관측된 실험 데이터를 분석하는 것**
  - 기존에 사용되는 대부분의 통계적 기법이 바로 적용되기 어려움
    - 기존에는 많은 경우 관측 데이터가 독립적이고 동일한 분포를 가진다고 가정 (independent and identically distributed, i.i.d.)
    - 하지만 **시계열 데이터는 인접하여 관측된 데이터의 경우 분명한 관계성**이 있음
  - 따라서 시계열 분석에서는 이러한 부분을 잘 다룰 수 있어야 함

가장 오래 된 시계열 차트

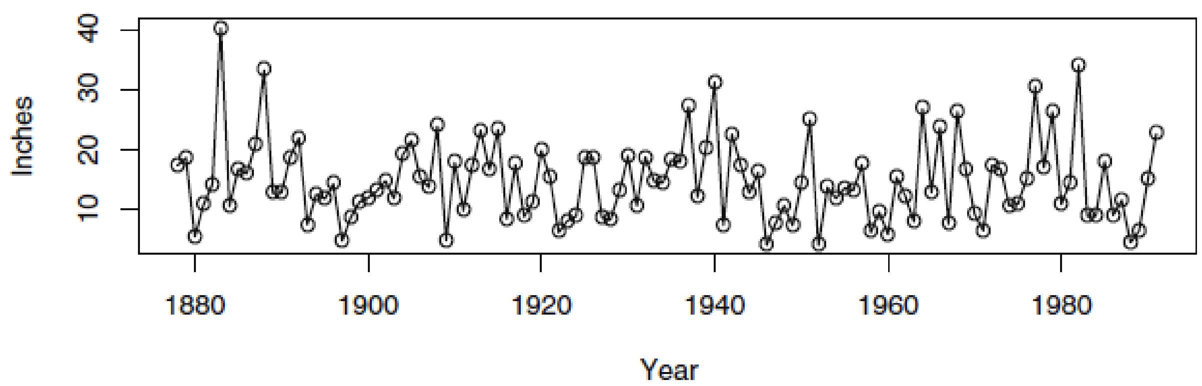
Exhibit 1.10 A Tenth-Century Time Series Plot



- 약 10, 11세기에 그려진 것으로 알려진 천체의 움직임

예시 1: Los Angeles의 연간 강우량

Exhibit 1.1 Time Series Plot of Los Angeles Annual Rainfall

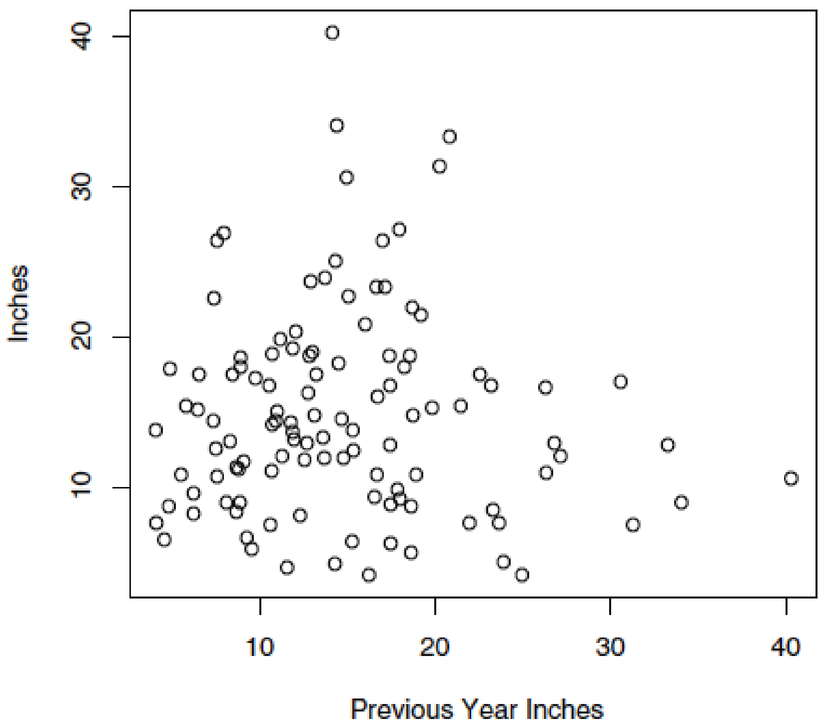


- **간단한 관측**
  - 매년 강우량 변동이 꽤 크다
  - 1880년대에 아주 큰 강우량을 기록한 적이 있다

- **간단한 질문**
  - 전해와 이듬해의 강우량에 관계성이 있을까?

예시 1: Los Angeles의 연간 강우량

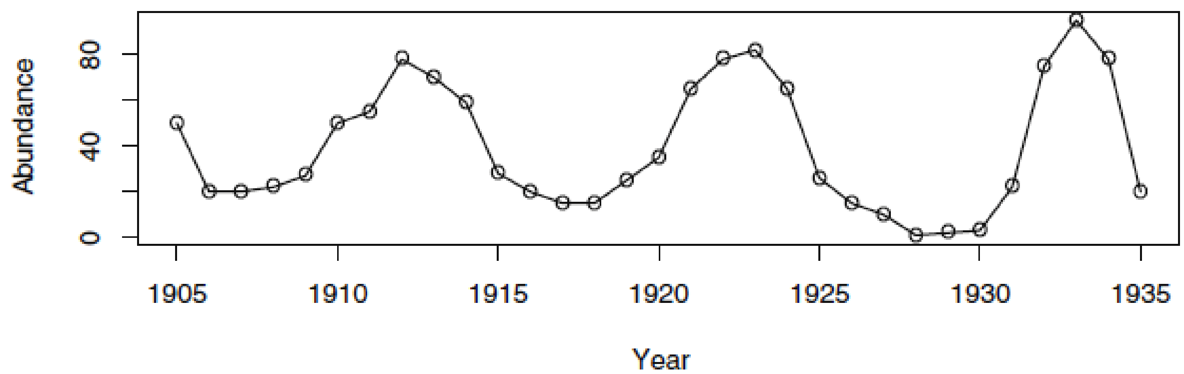
Exhibit 1.2 Scatterplot of LA Rainfall versus Last Year's LA Rainfall



- **간단한 질문**
  - 전해와 이듬해의 강우량에 관계성이 있을까? **아마 아닐 것**

예시 2: 연간 토끼 개체 수

Exhibit 1.5 Abundance of Canadian Hare

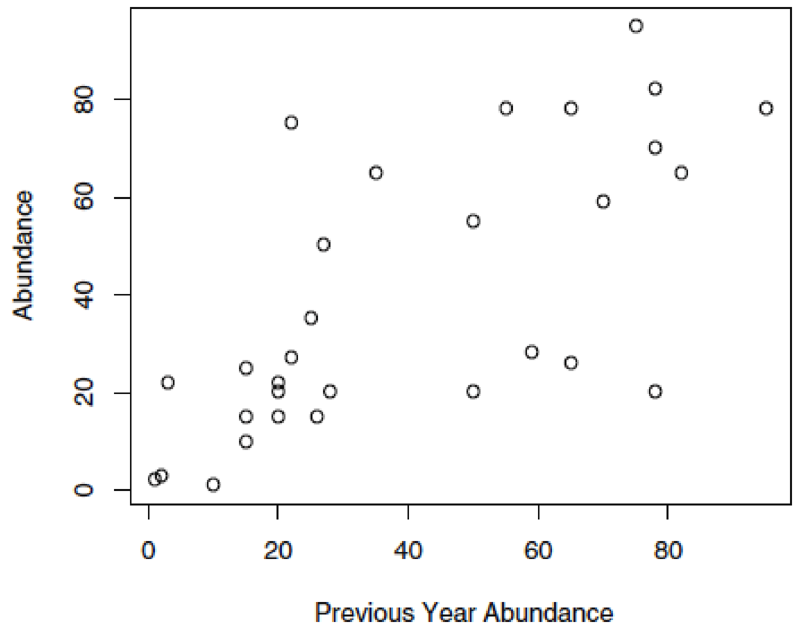


- **간단한 관측**
  - 인접한 관측 값들이 유사한 값을 가짐

- **간단한 질문**
  - 전해와 이듬해의 개체 수에 관계성이 있을까?

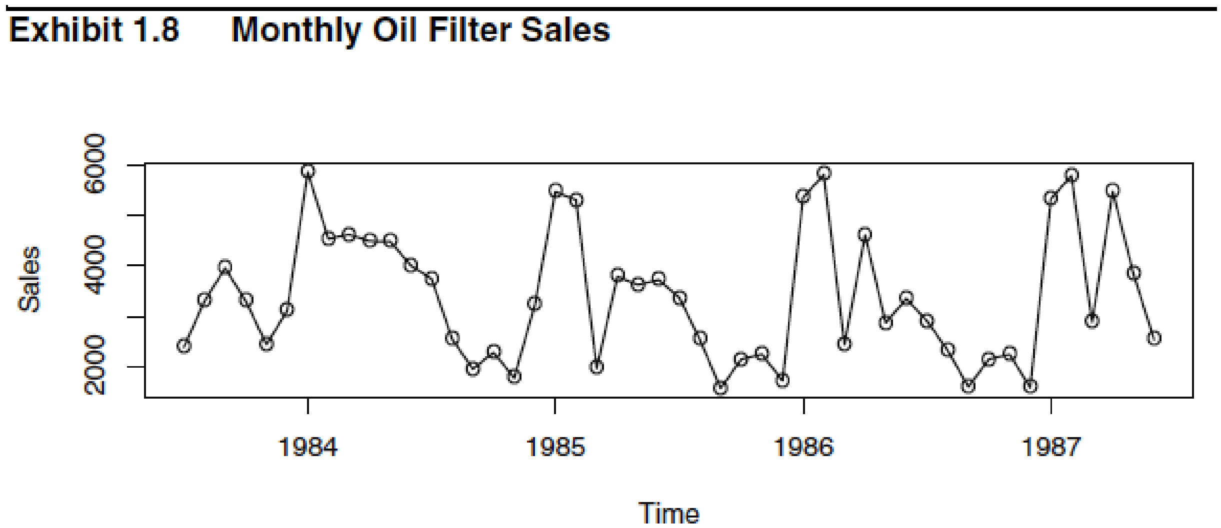
예시 2: 연간 토끼 개체 수

Exhibit 1.6 Hare Abundance versus Previous Year's Hare Abundance



- **간단한 질문**
  - 전해와 이듬해의 개체 수에 관계성이 있을까? 그런 듯

예시 3: 월간 오일 필터 판매량

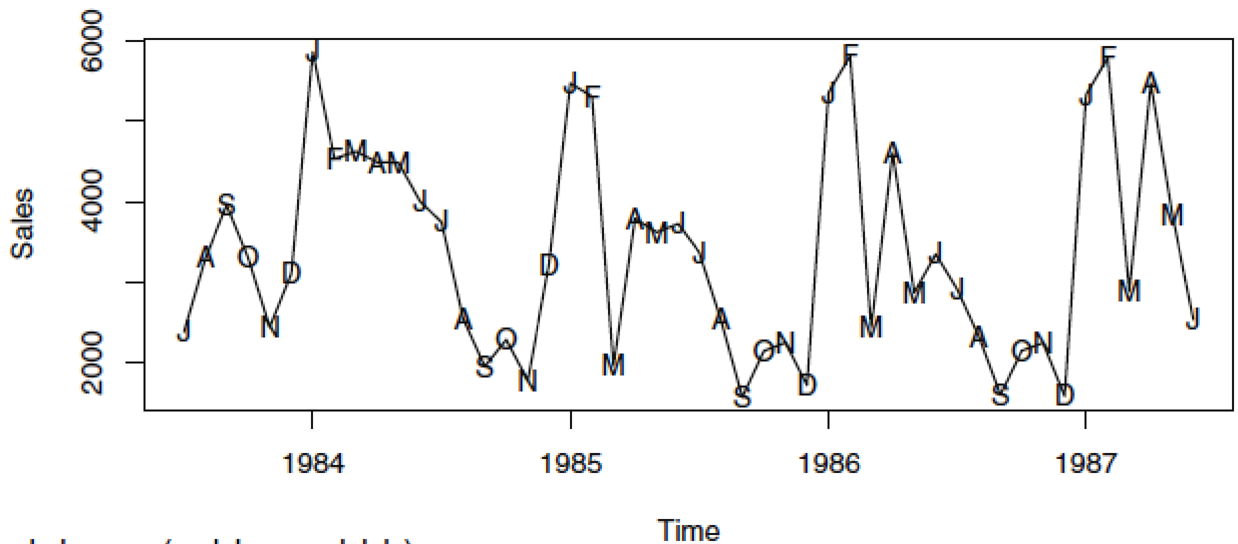


- **간단한 관측**
  – 상당히 변동 폭이 큰 듯

- **간단한 질문**
  – 데이터에 계절성(seasonality)이 있을까?

예시 3: 월간 오일 필터 판매량

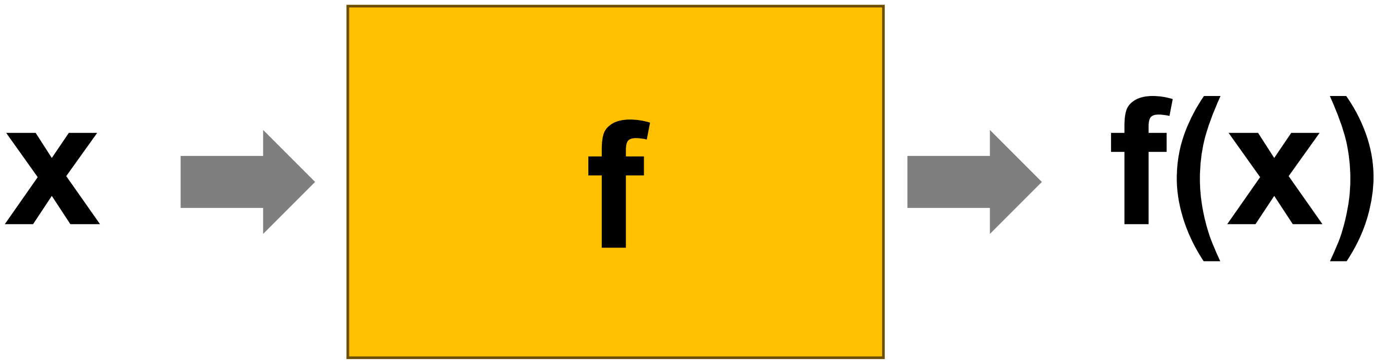
Exhibit 1.9 Monthly Oil Filter Sales with Special Plotting Symbols



J=January (and June and July),
F=February, M=March (and May), and so forth

- **간단한 질문**
  - – 데이터에 계절성(seasonality)이 있을까? **그런 듯**

예측 모형의 원리



예측 모형의 원리

x → **ML/AI** → **f(x)**

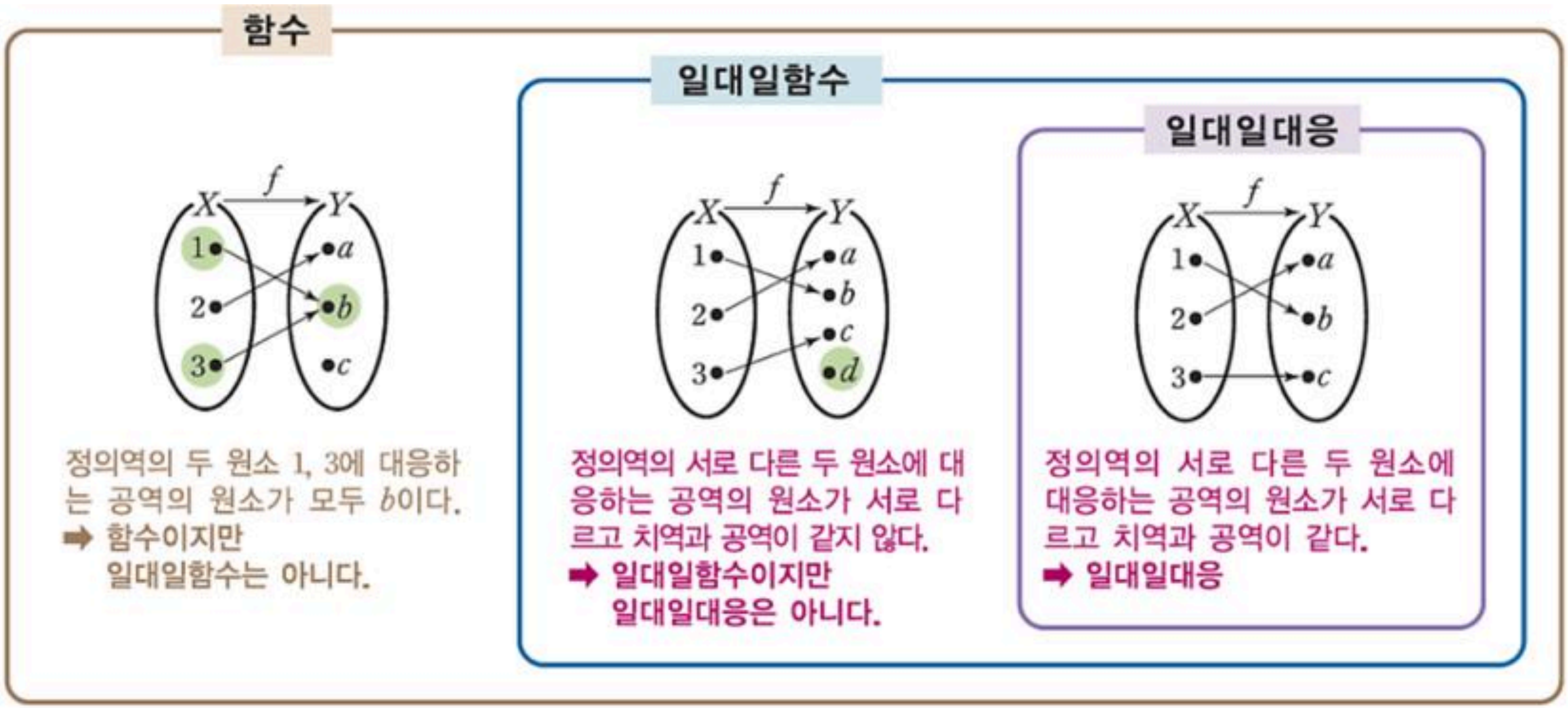
예측 모형의 원리

ML/AI

Dog

예측 모형의 원리

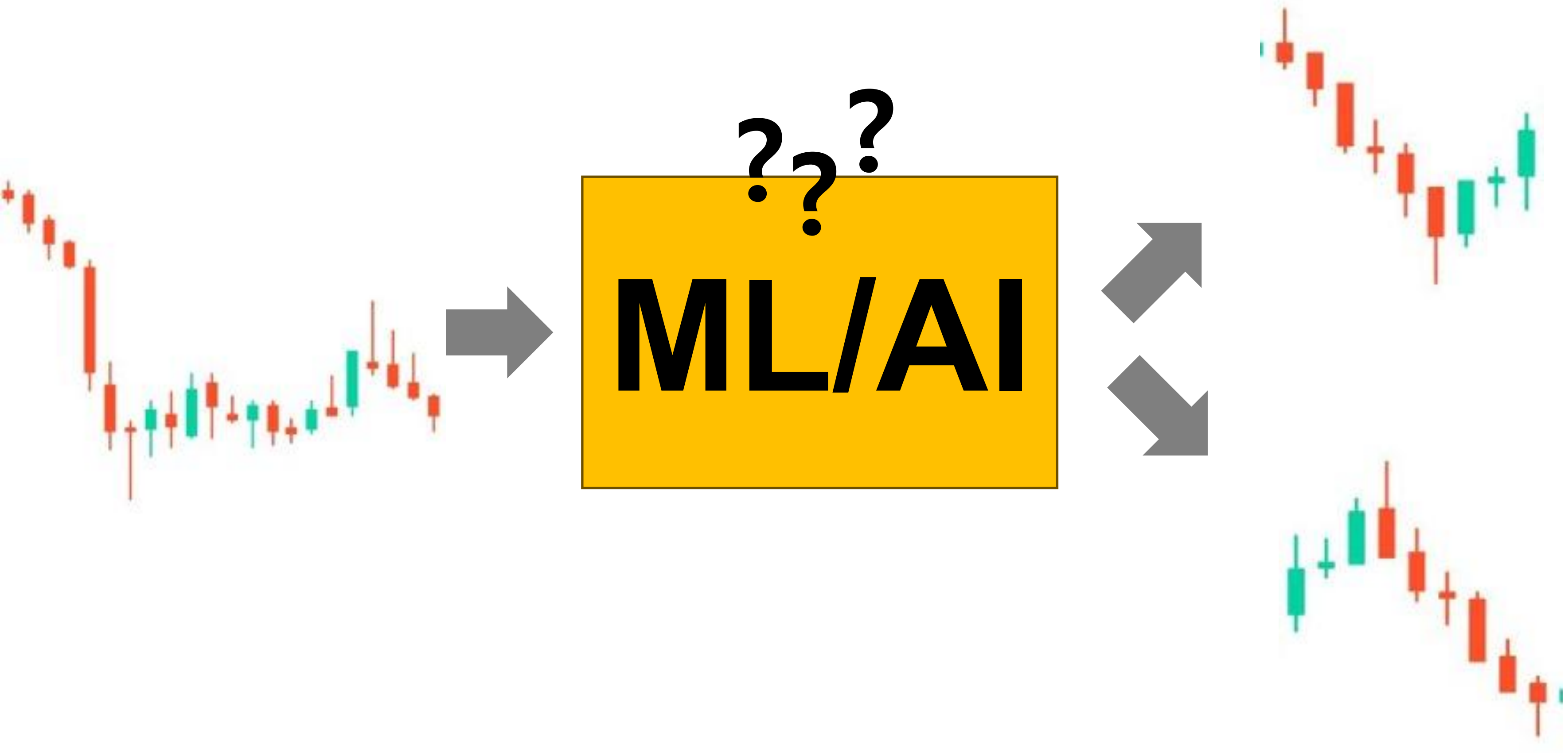
예측 모형의 원리



함수
일대일함수
일대일대응
f
X
Y
1
2
3
a
b
c
d
정의역의 두 원소 1, 3에 대응하는 공역의 원소가 모두 b이다.
➡ 함수이지만
일대일함수는 아니다.
정의역의 서로 다른 두 원소에 대응하는 공역의 원소가 서로 다르고 치역과 공역이 같지 않다.
➡ 일대일함수이지만
일대일대응은 아니다.
정의역의 서로 다른 두 원소에 대응하는 공역의 원소가 서로 다르고 치역과 공역이 같다.
➡ 일대일대응

예측 모형의 원리

예측 모형의 원리



예측 모형의 원리

조금 거칠게 말해서 함수의 성질이 잘 지켜는 데이터
즉, 하나의 입력값에 하나의 출력값이 잘 대응되는 데이터는
정상성(stationarity)이 있는 데이터라고 할 수 있으며,
이러한 데이터는 예측이 잘 되는 편

예측 모형의 원리

x →  → f(x)

예측 모형의 원리

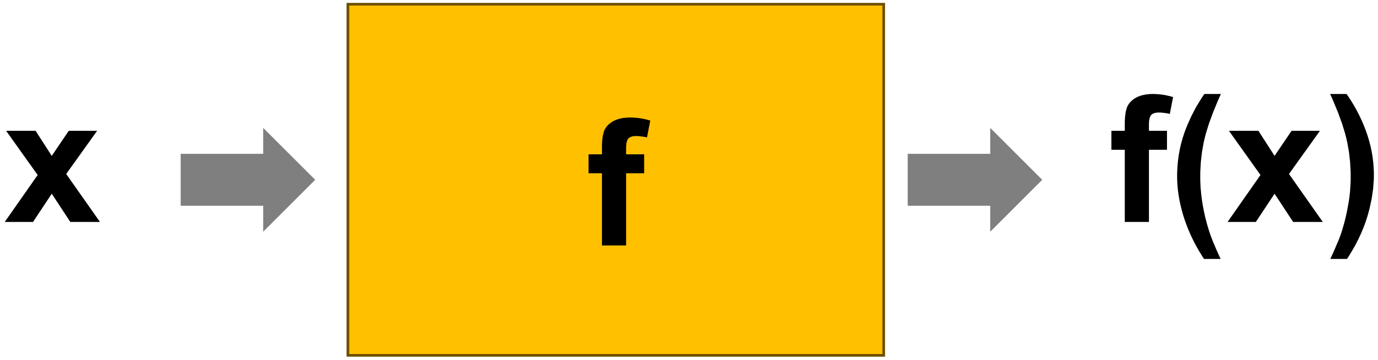
미국 대선은
누가 이겼니?

도널드 트럼프

Model-driven 방식

- 전통적 기법들은 대부분 model-driven이라 볼 수 있음
  - 계산되는 수식이 큰 틀에서 이미 결정되어있음
  - 데이터 분포에 대한 가정이 있는 경우가 많음
  - 장점
    - 도메인 지식을 활용하기 쉬움
    - 분석 결과를 일반화 하기 쉬움
  - 단점
    - 데이터나 분석 환경이 복잡할 경우엔 사용하기 어려움

Model-driven 방식



Ex)

ARIMA

$$x'_t = b_0 + b_1 x'_{t-1} + \ldots + b_p x'_{t-p} + \theta_1 e_{t-1} + \ldots + \theta_q e_{t-q} + e_t$$

DIFFERENCED 'x' TIME SERIES (d) PART

AR (p) PART

MA (q) PART

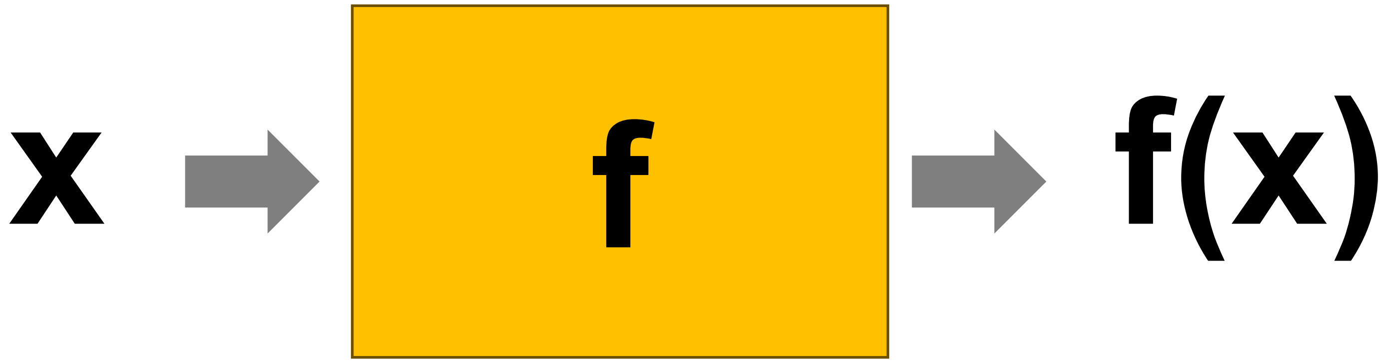
GARCH

$$\sigma_t^2 = \omega + \underbrace{\sum_{i=1}^{q} \alpha_i \epsilon_{t-i}^2}_{\text{ARCH term}} + \underbrace{\sum_{j=1}^{p} \beta_j \sigma_{t-j}^2}_{\text{GARCH term}}$$

Data-driven 방식

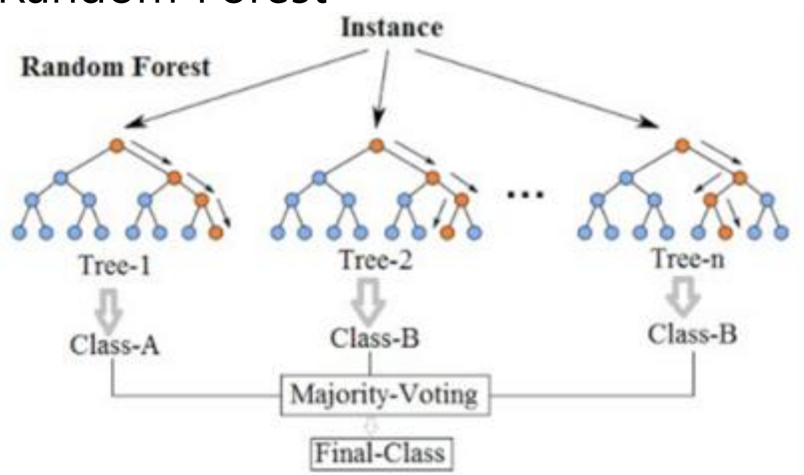
- AI 기법은 대부분 data-driven 방식으로 볼 수 있음
  - 계산 수식이나 데이터의 확률 분포에 대해 특별한 가정이 없음
  - 장점
    - 다양한 변수 사이의 복잡한 (일반적으로 비선형적인) 관계성을 반영할 수 있음
  - 단점
    - 충분한 양의 데이터가 필요
    - 결과를 해석하기가 다소 어려움

Data-driven 방식

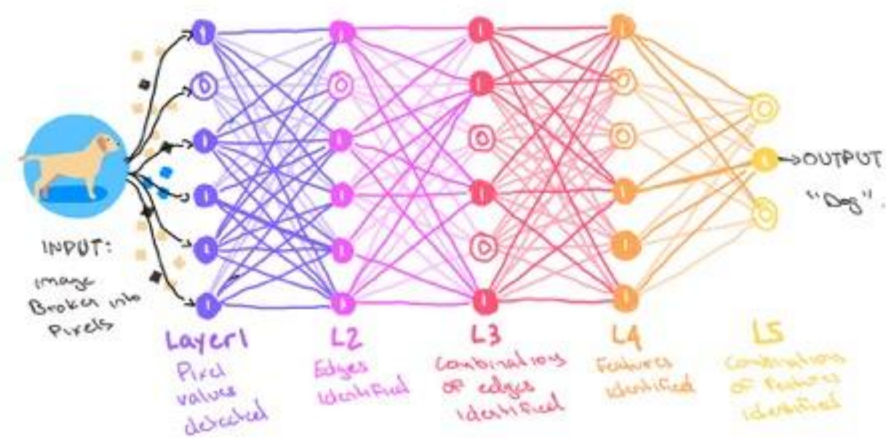


Ex)

Random Forest



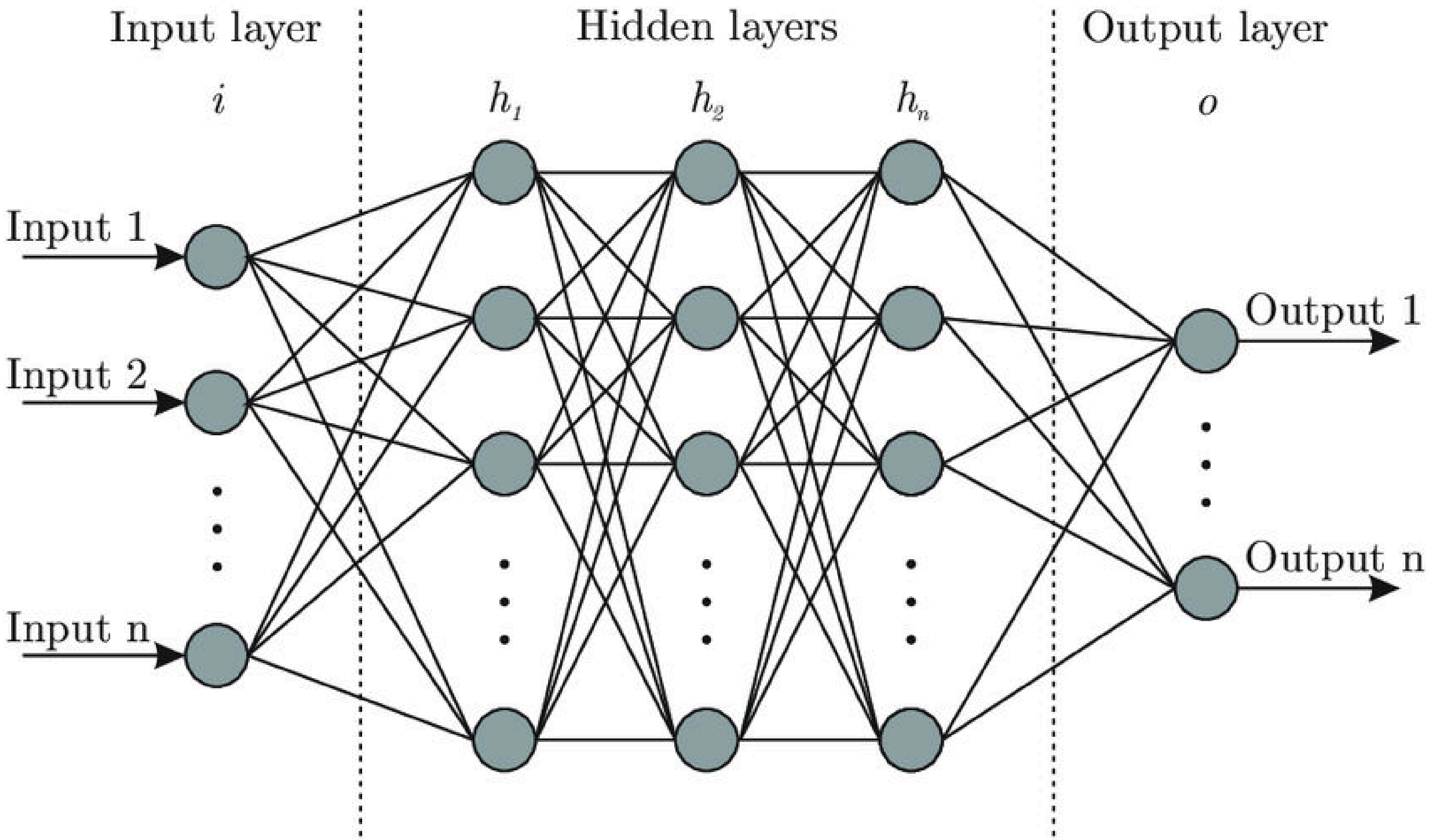
Neural Network



Section 2

Deep Learning Models for Time Series Analysis

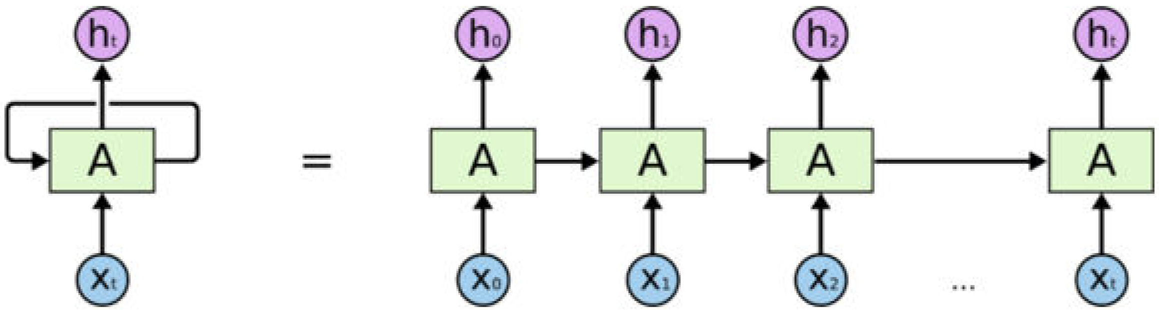
Artificial neural networks



2.1

RNN-based models

Recurrent neural networks (RNN)

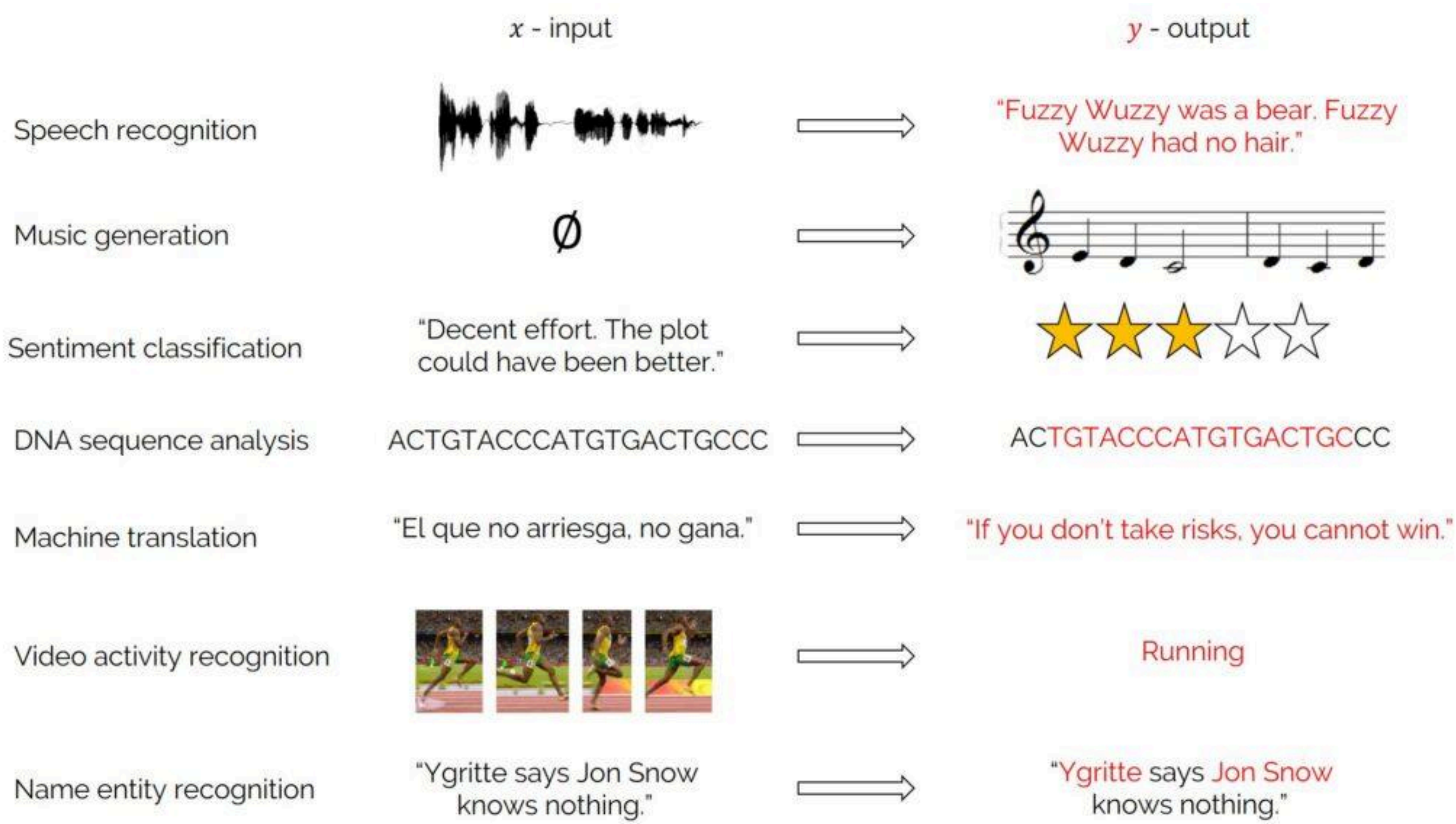


An unrolled recurrent neural network.

http://colah.github.io/posts/2015-08-Understanding-LSTMs/

Recurrent neural networks (RNN)

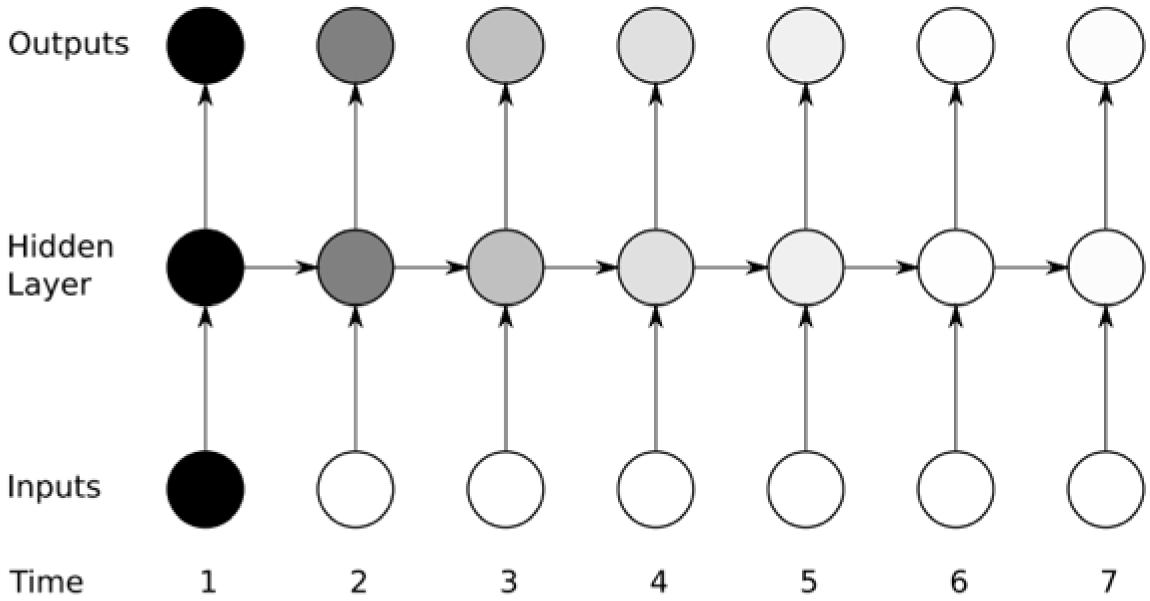
- RNNs are widely used in



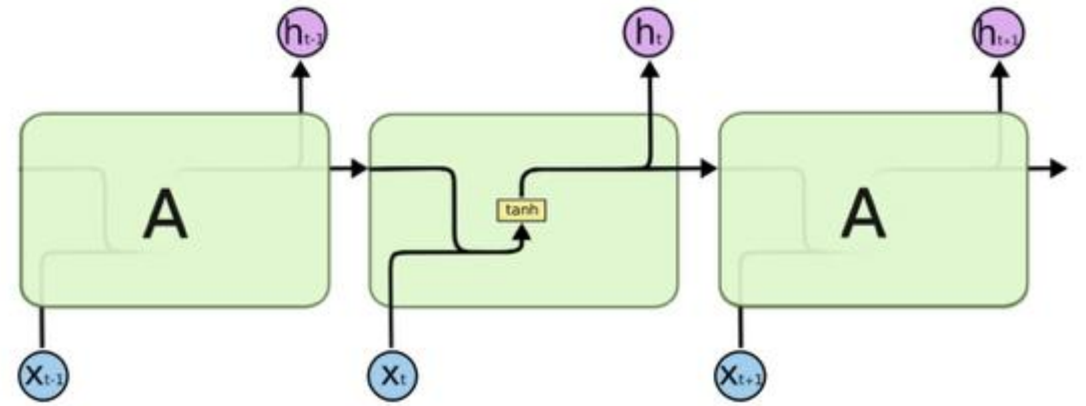
http://datahacker.rs/003-rnn-architectural-types-of-different-recurrent-neural-networks/

Vanishing gradient of RNNs

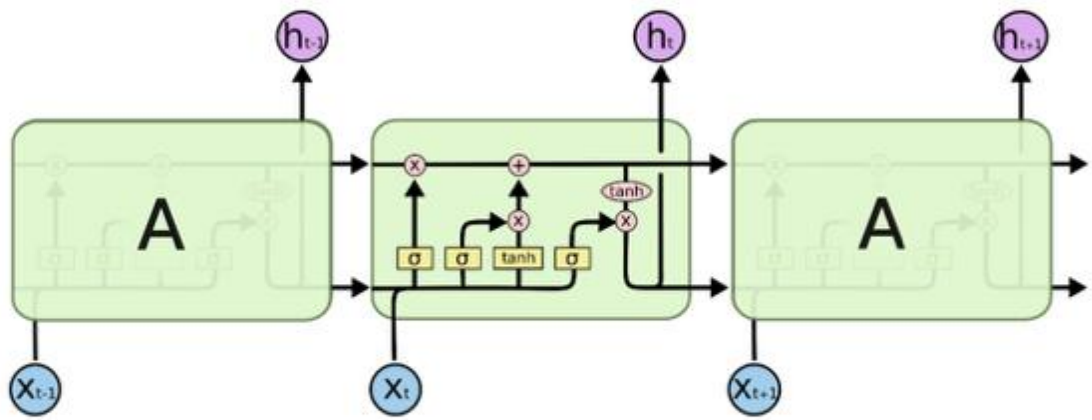
- 기본 RNN 구조의 한계점
  - 같은 구조를 계속해서 반복하기 때문에 input의 영향이 뒤로 갈 수록 점점 미미해지거나 아니면 오히려 너무 지나치게 증폭 될 수 있음
  - 간단하게 말하자면, RNN이 초기 입력값을 ***잊어버림***



Long short term memory (LSTM)



The repeating module in a standard RNN contains a single layer.



The repeating module in an LSTM contains four interacting layers.

http://colah.github.io/posts/2015-08-Understanding-LSTMs/

Neural ODE

- RNN과 LSTM 모두 관측값이 동일한 간격으로 관측되었다고 가정

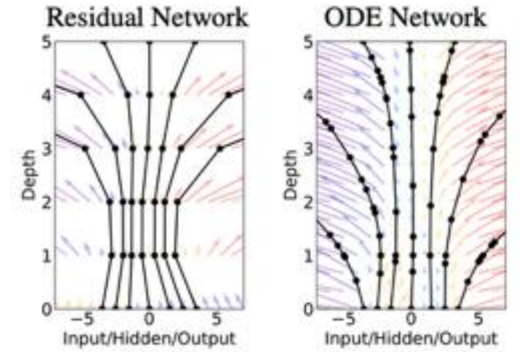


Figure 1: *Left:* A Residual network defines a discrete sequence of finite transformations. *Right:* A ODE network defines a vector field, which continuously transforms the state. *Both:* Circles represent evaluation locations.

- NeuralODE (Chen et al., 2018)는 일정하지 않게 관측된 데이터도 다룰 수 있도록 hidden state를 continuous하게 설정
  - ANN은 hidden state의 **derivative** (미분값, 변화량)을 학습

$$\mathbf{h}_{t+1} = \mathbf{h}_t + f(\mathbf{h}_t, \theta_t)$$

$$\frac{d\mathbf{h}(t)}{dt} = f(\mathbf{h}(t), t, \theta)$$

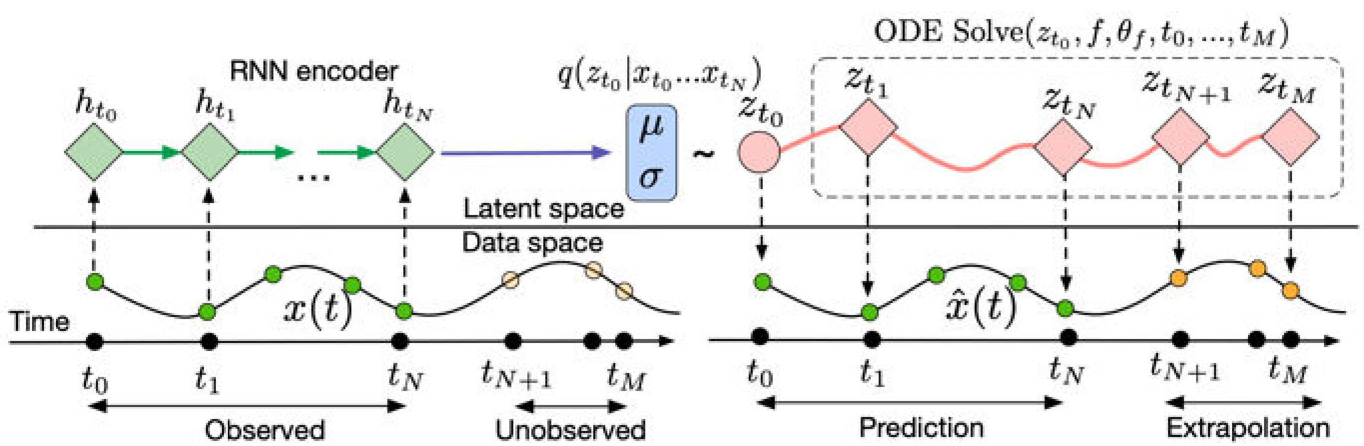


Figure 6: Computation graph of the latent ODE model.

RNN-based models

- Example: DeepAR (Salinas et al., 2020) by Amazon

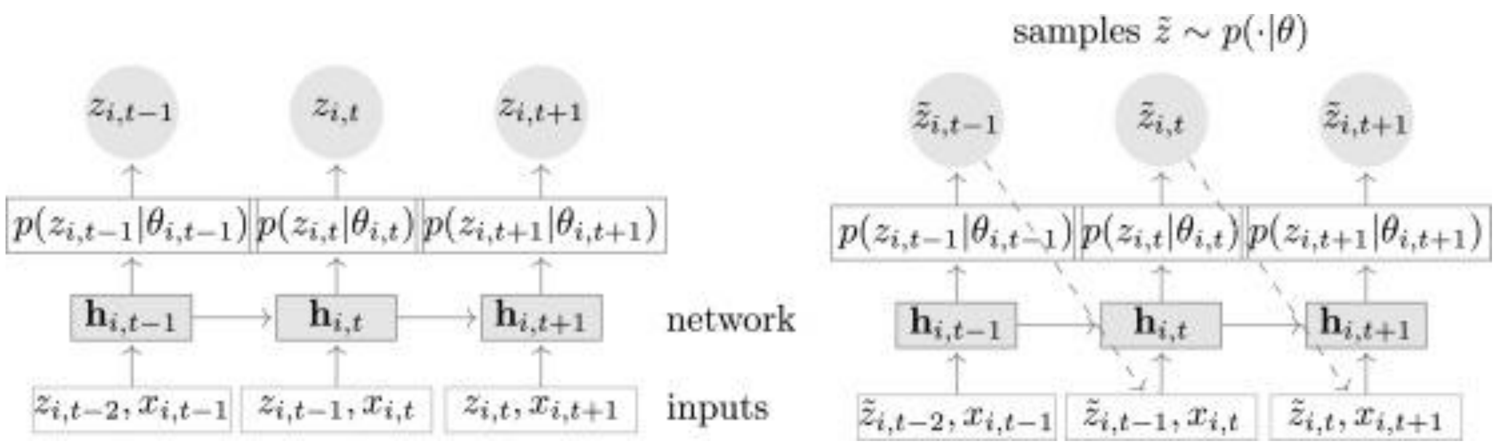


Fig. 3. Summary of the model. Training (left): At each time step t, the inputs to the network are the covariates $x_{i,t}$, the target value at the previous time step $z_{i,t-1}$, and the previous network output $\mathbf{h}_{i,t-1}$. The network output $\mathbf{h}_{i,t} = h\left(\mathbf{h}_{i,t-1}, z_{i,t-1}, \mathbf{x}_{i,t}, \Theta\right)$ is then used to compute the parameters $\theta_{i,t} = \theta\left(\mathbf{h}_{i,t}, \Theta\right)$ of the likelihood $p\left(z|\theta\right)$, which is used for training the model parameters. For prediction, the history of the time series $z_{i,t}$ is fed in for $t < t_0$, then in the prediction range (right) for $t \geq t_0$ a sample $\hat{z}_{i,t} \sim p\left(\cdot|\theta_{i,t}\right)$ is drawn and fed back for the next point until the end of the prediction range $t = t_0 + T$, generating one sample trace. Repeating this prediction process yields many traces that represent the joint predicted distribution.

https://doi.org/10.1016/j.ijforecast.2019.07.001

RNN-based models

- Example: DeepAR (Salinas et al., 2020) by Amazon

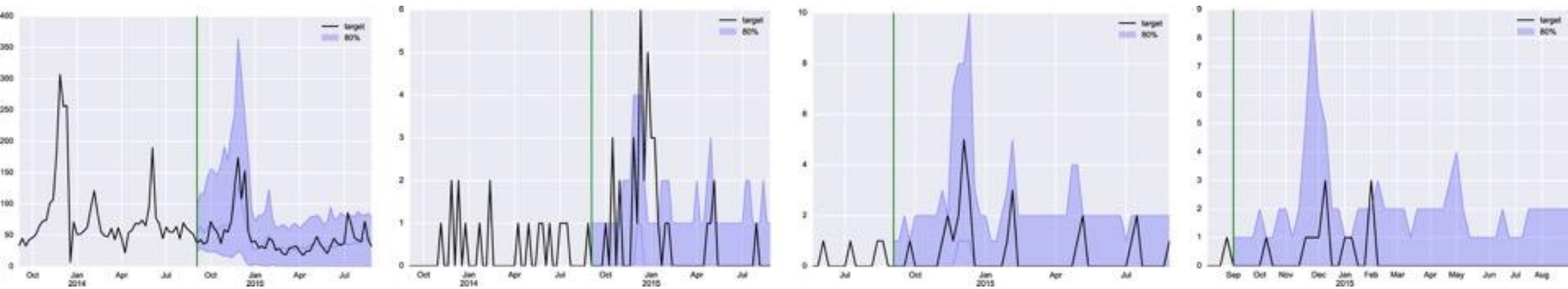


Fig. 5. Example time series of ec. The vertical line separates the conditioning period from the prediction period. The black line shows the true target. In the prediction range, we plot the p50 as a blue line (mostly zero for the three slow items), along with 80% confidence intervals (shaded). The model learns accurate seasonality patterns and uncertainty estimates for items of different velocities and ages . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

https://doi.org/10.1016/j.ijforecast.2019.07.001

2.2

Generative models

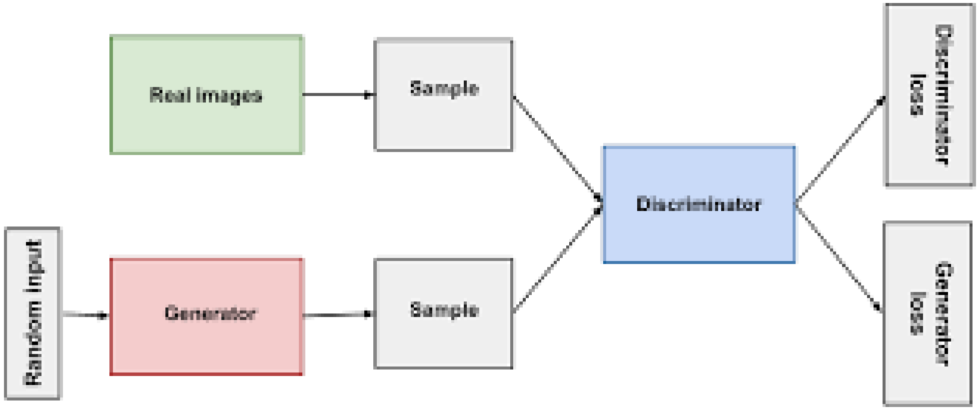
Generative models

- **Generative** models vs. **discriminative** models
  - 간단하게 말하자면
    - **Discriminative** models 데이터의 종류를 판별하는 모형
      - 예를 들면 사진을 바탕으로 개인지 고양이인지 구분
    - **Generative** models은 새로운 데이터를 만들어내는 모형
      - 예를 들어 여러 이미지를 학습 후 새로운 이미지를 만들어냄

Source: Google Developers

Generative Adversarial Networks (GAN)

- Goodfellow et al. (2014)가 제안한 두 네트워크가 게임을 하듯이 경쟁하는 생성모형
  - Generator (generative network)
    - 데이터 분포를 학습하여 새로운 데이터를 만들어내고자 함
    - Discriminator를 속이는 것이 목표
  - Discriminator (discriminative network)
    - Generator가 생성한 가짜 데이터와 진짜 데이터를 구분하는 것이 목표

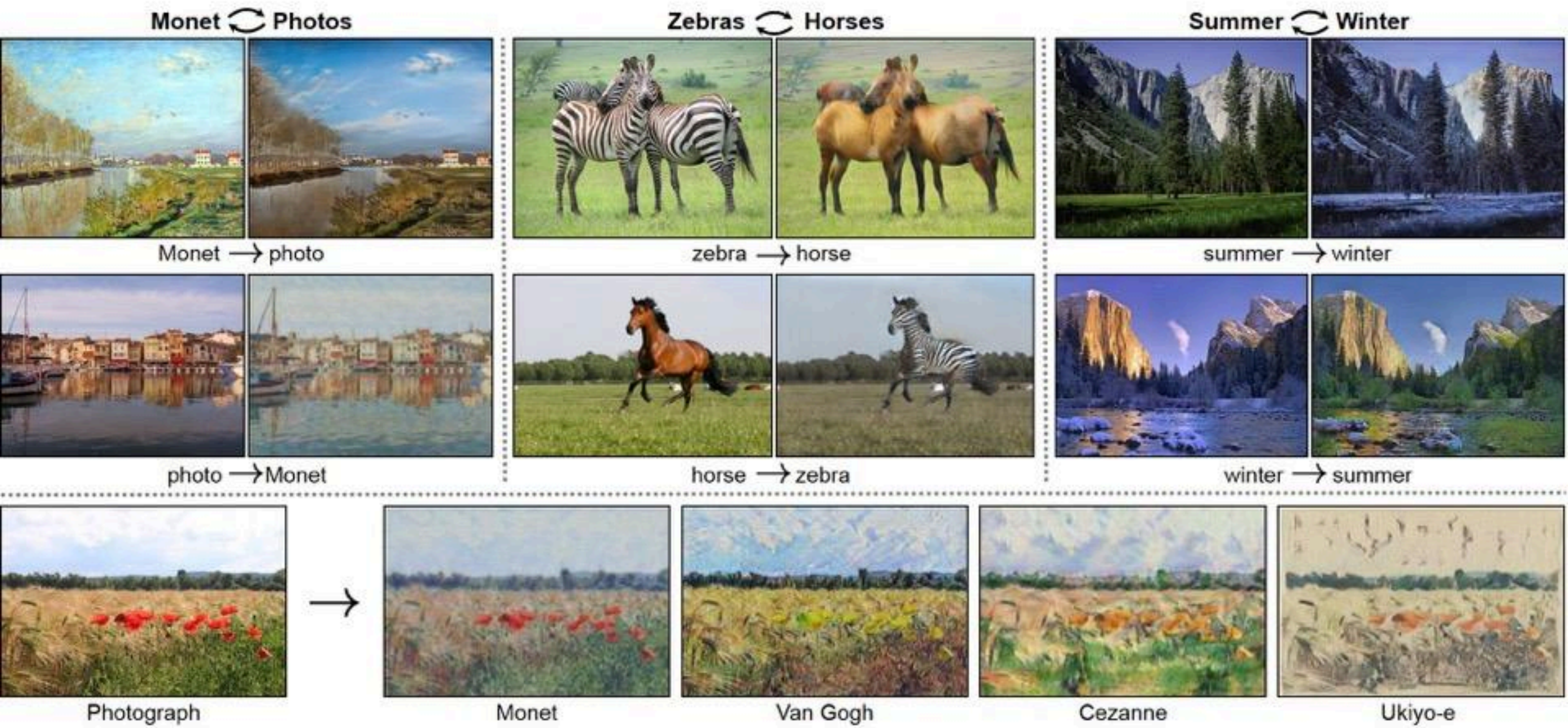


Generative Adversarial Networks (GAN)

- Examples: StyleGAN (Karras et al., 2019) by NVIDIA

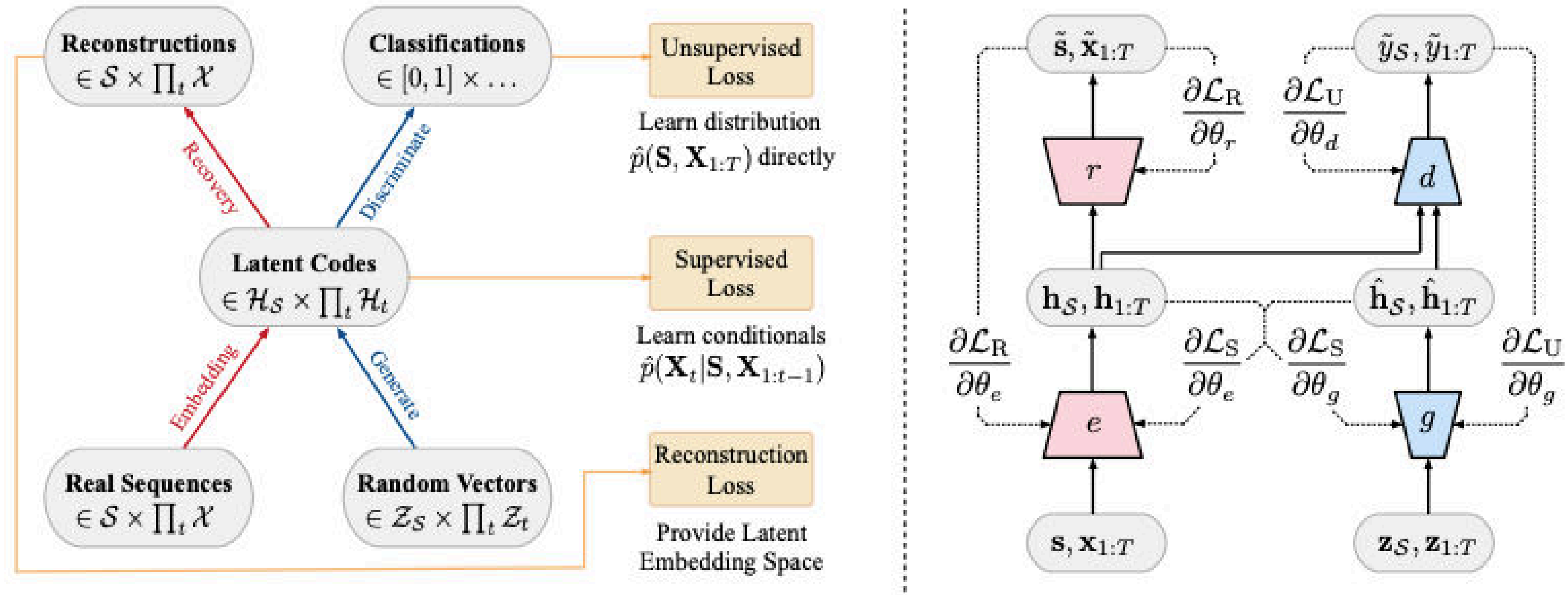
Generative Adversarial Networks (GAN)

- Examples: CycleGAN (Zhu et al., 2017)



Generative Adversarial Networks (GAN)

- Examples: TimeGAN (Yoon et al., 2019)



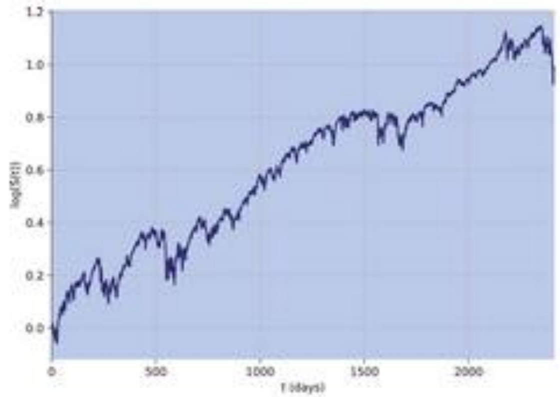
(a) Block Diagram (b) Training Scheme

Figure 1: (a) Block diagram of component functions and objectives. (b) Training scheme; solid lines indicate forward propagation of data, and dashed lines indicate backpropagation of gradients.

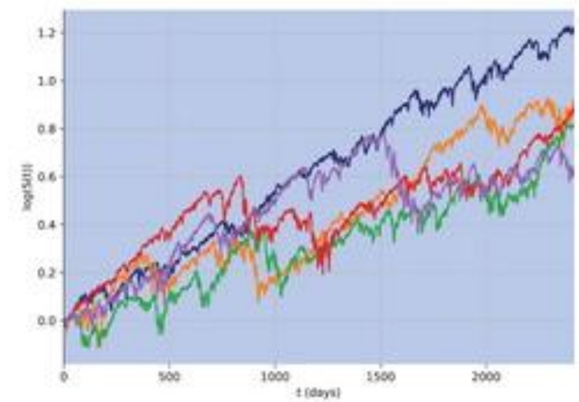
https://papers.nips.cc/paper/2019/hash/c9efe5f26cd17ba6216bbe2a7d26d490-Abstract.html

Generative Adversarial Networks (GAN)

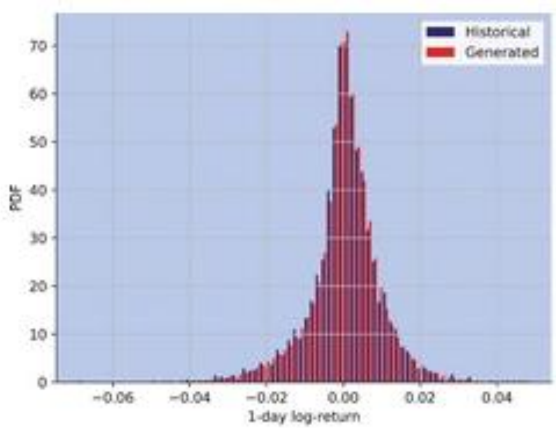
- Examples: QuantGANs (Wiese et al., 2020)



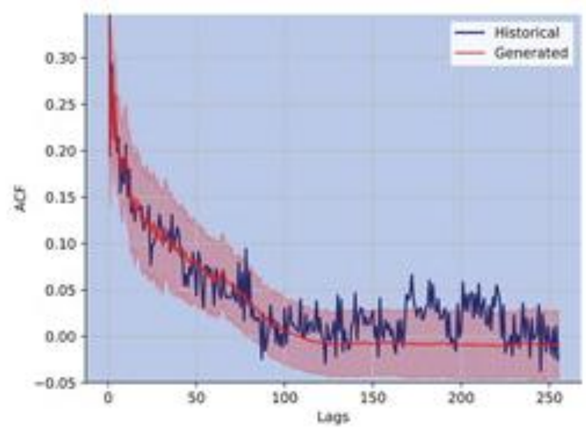
(a) S&P 500 log-path



(b) Generated log-paths of a Quant GAN



(a) Histogram of daily log-returns



(b) ACF of absolute log-returns

•https://doi.org/10.1080/14697688.2020.1730426

Generative Adversarial Networks (GAN)

- Examples: TadGAN (Geiger et al., 2020)

Fig. 1. An illustration of time series anomaly detection using unsupervised learning. Given a multivariate time series, the goal is to find out a set of anomalous time segments that have unusual values and do not follow the expected temporal patterns.

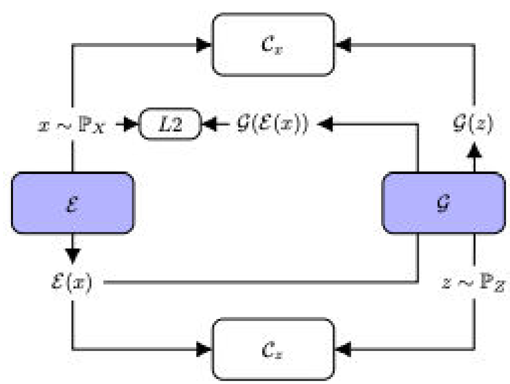


Fig. 2. Model architecture: *Generator* $\mathcal{E}$ is serving as an Encoder which maps the time series sequences into the latent space, while *Generator* $\mathcal{G}$ is serving as a Decoder that transforms the latent space into the reconstructed time series. *Critic* $\mathcal{C}_x$ is to distinguish between real time series sequences from X and the generated time series sequences from $\mathcal{G}(z)$, whereas *Critic* $\mathcal{C}_z$ measures the goodness of the mapping into the latent space.

https://ieeexplore.ieee.org/abstract/document/9378139

Diffusion models

- 데이터에 점점 노이즈를 더하면 언젠가는 완전한 노이즈가 됨
- 그 반대 과정을 학습할 수 있다면 노이즈로부터 실제 같은 데이터를 생성해낼 수 있음

Image source: S. Orbell's blog

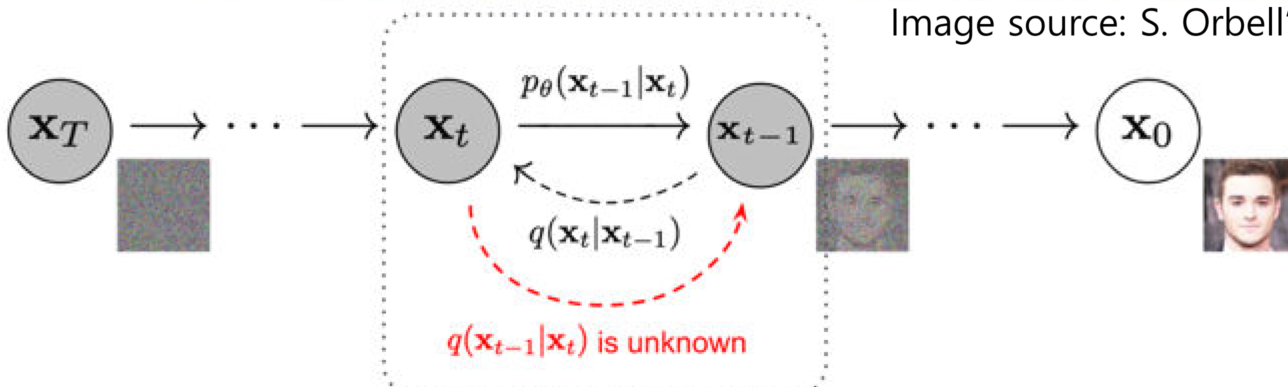


Image source: Ho et al. (2020)

Diffusion models

- Example: TimeGrad (Rasul et al., 2021)

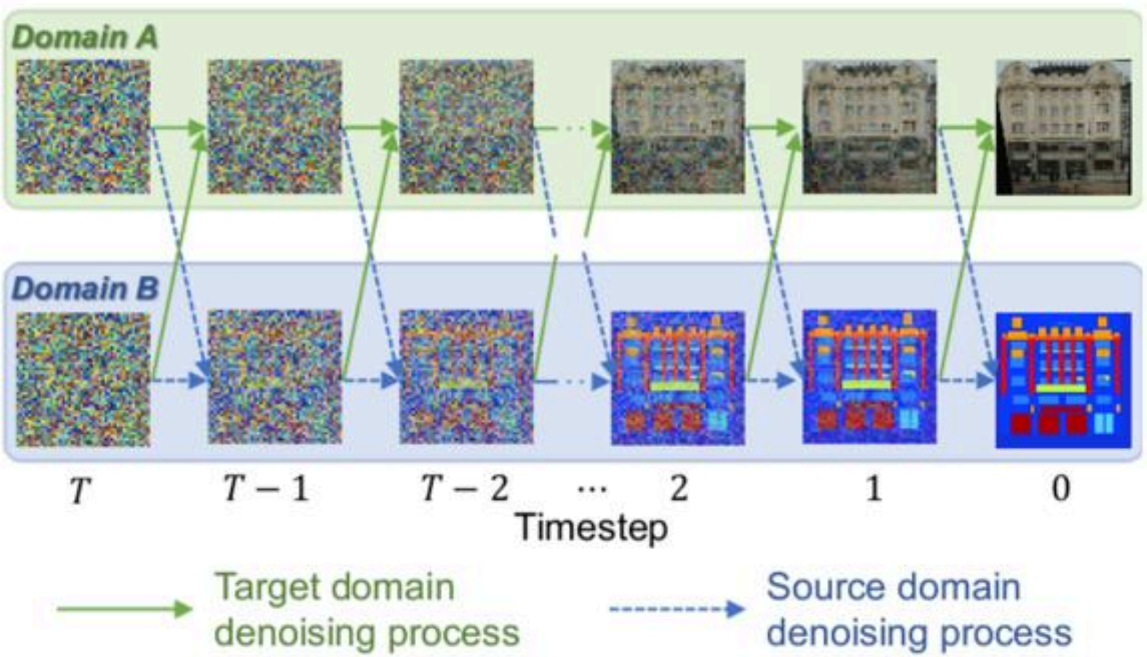


Figure 1: Conceptual illustration of our novel image-to-image translation approach using denoising diffusion probabilistic models.

https://proceedings.mlr.press/v139/rasul21a.html

Diffusion models

- Example: TimeGrad (Rasul et al., 2021)

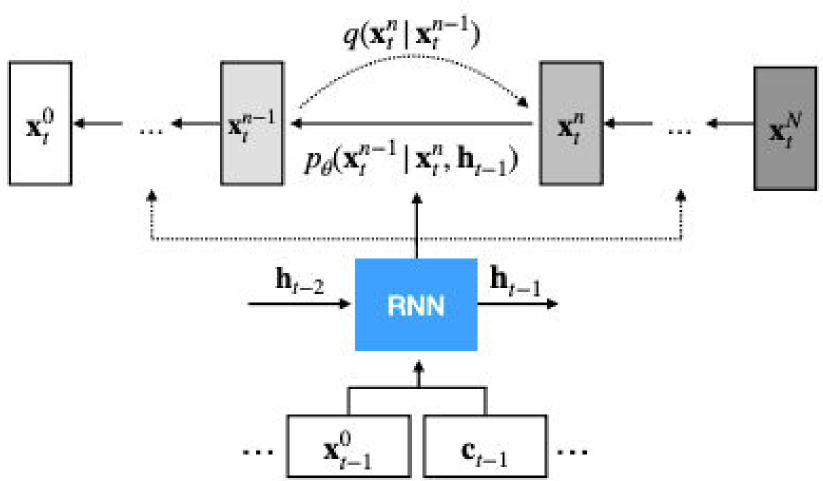


Figure 1. TimeGrad schematic: an RNN *conditioned* diffusion probabilistic model at some time $t-1$ depicting the fixed forward process that adds Gaussian noise and the learned reverse processes.

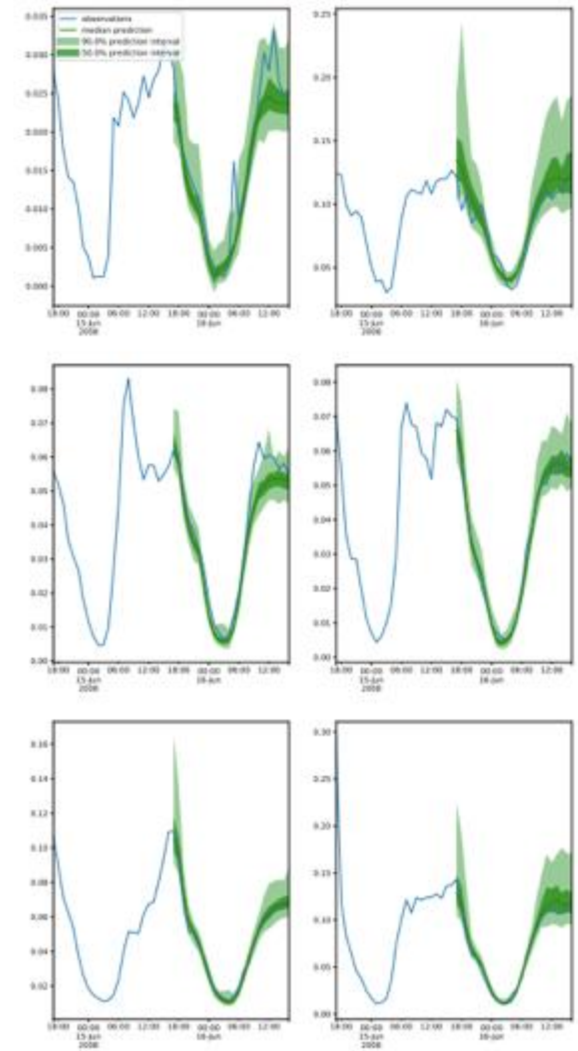


Figure 4. TimeGrad prediction intervals and test set ground-truth for Traffic data of the first 6 of 963 dimensions from first rolling-window. Note that neighboring entities have an order of magnitude difference in scales.

https://proceedings.mlr.press/v139/rasul21a.html

2.3

Attention-based models

Sequence-to-sequence models

- sequence-to-sequence (seq2seq) models에서는 input이 sequence로 들어가며 hidden state가 update되고, 이를 바탕으로 output을 sequence로 출력

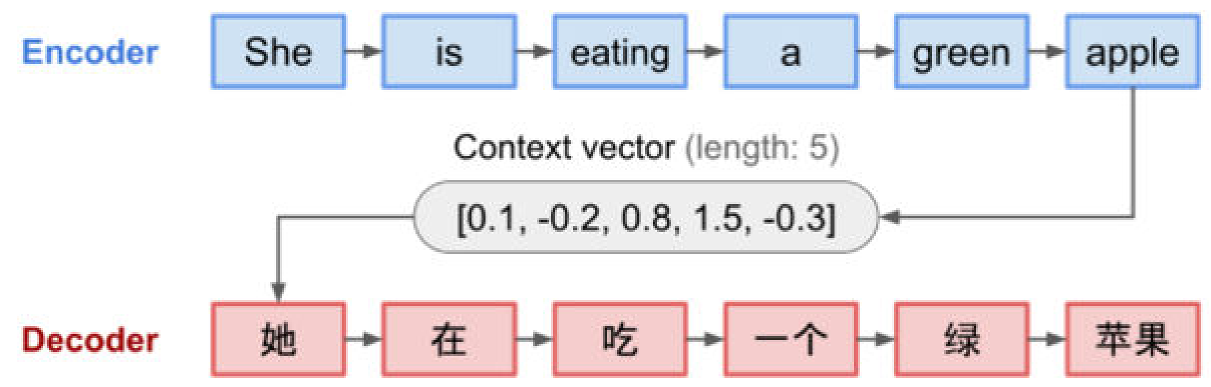


Fig. 3. The encoder-decoder model, translating the sentence "she is eating a green apple" to Chinese. The visualization of both encoder and decoder is unrolled in time.

Image source: Lil'Log

Sequence-to-sequence models

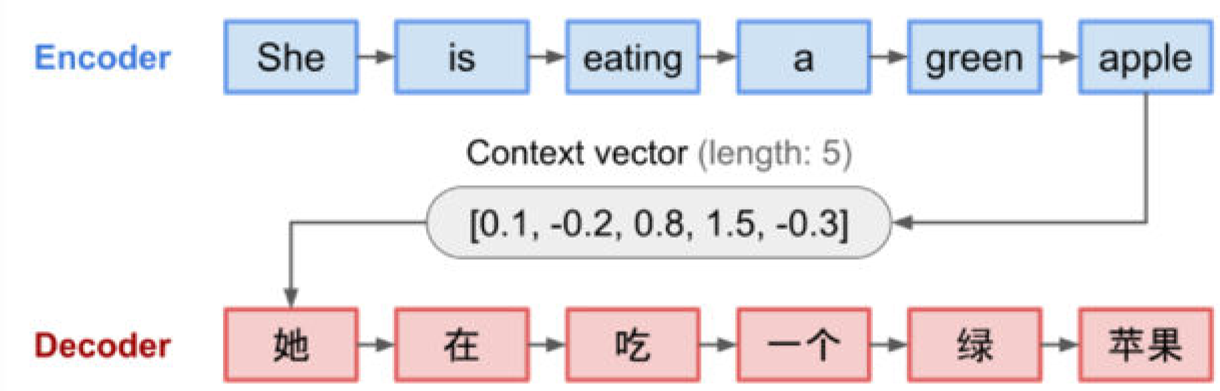


Fig. 3. The encoder-decoder model, translating the sentence "she is eating a green apple" to Chinese. The visualization of both encoder and decoder is unrolled in time.

Image source: Lil'Log

- Input이 길어지게 되면 이를 기억하기가 어려움

Attention mechanism

- 따라서 전체 input을 하나로 요약하기보다, input에서 어떤 부분을 더 '집중(attend)'해야하는지 판단하여 사용

Fig. 2. One word "attends" to other words in the same sentence differently.

Image source: Lil'Log

Attention mechanism

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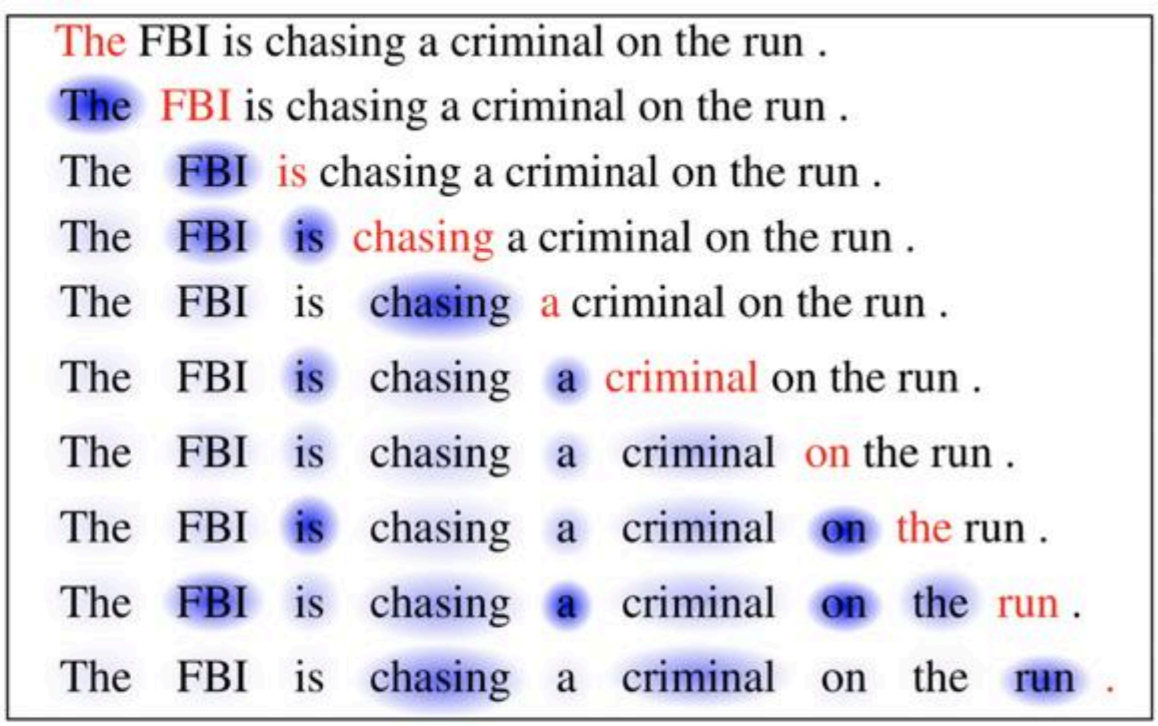


Image source: Cheng et al. (2016)

Attention is all you need (Transformers)

- Vaswani et al. (2017)는 RNN cell 없이 attention만으로 구성된 Transformer 모형을 제안
  - 여러 task에서 매우 높은 성능을 보임
  - 가장 큰 장점은 병렬화 시켜 모델을 매우 크게 만들기가 용이하여, 엄청나게 많은 양의 데이터를 학습하는 것이 가능해짐

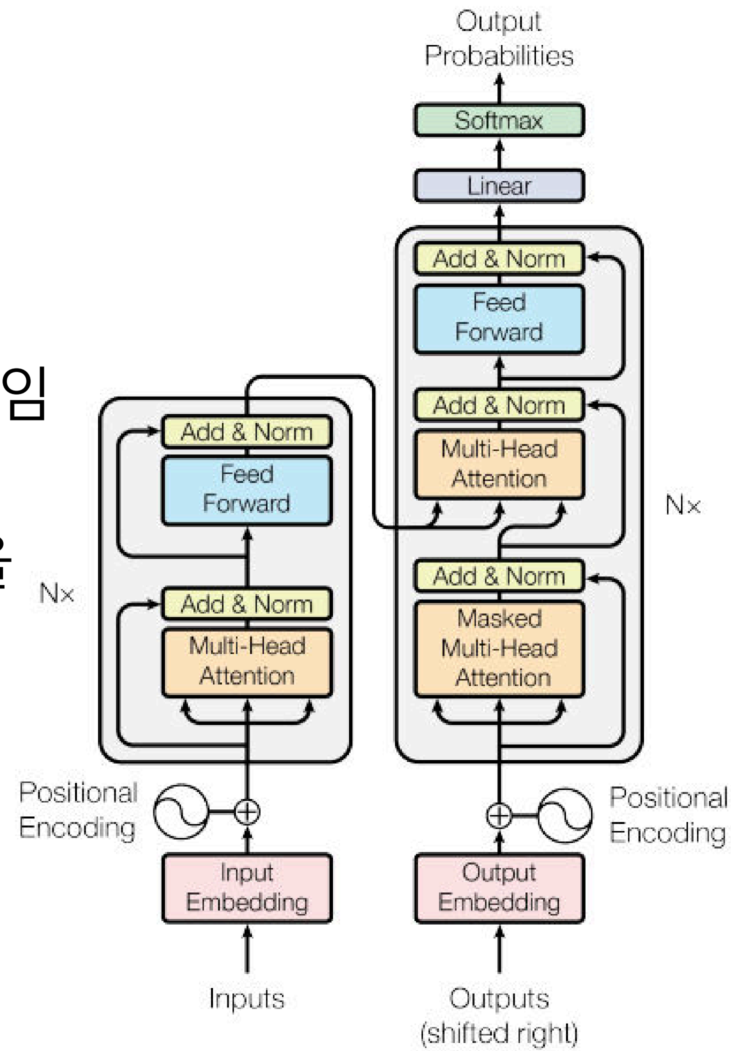
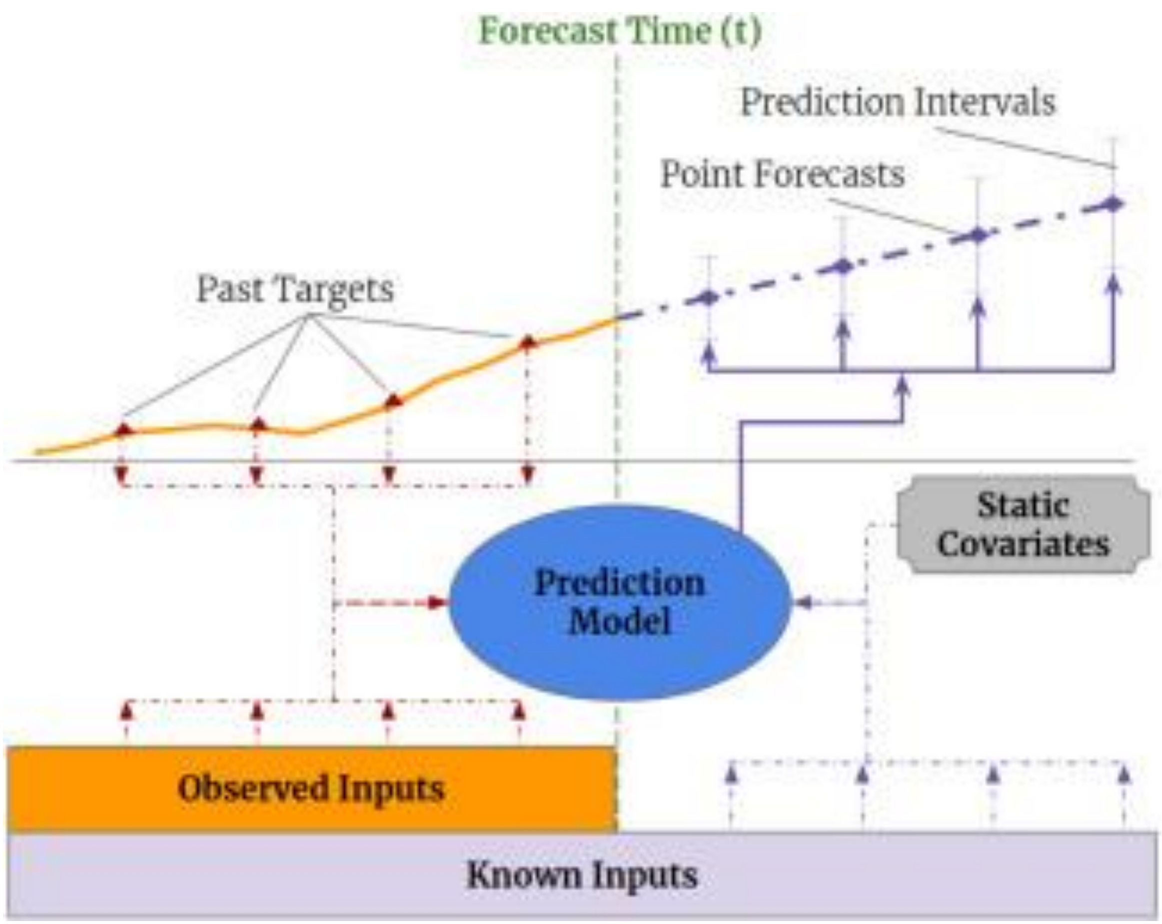


Figure 1: The Transformer - model architecture.

Attention is all you need (Transformers)

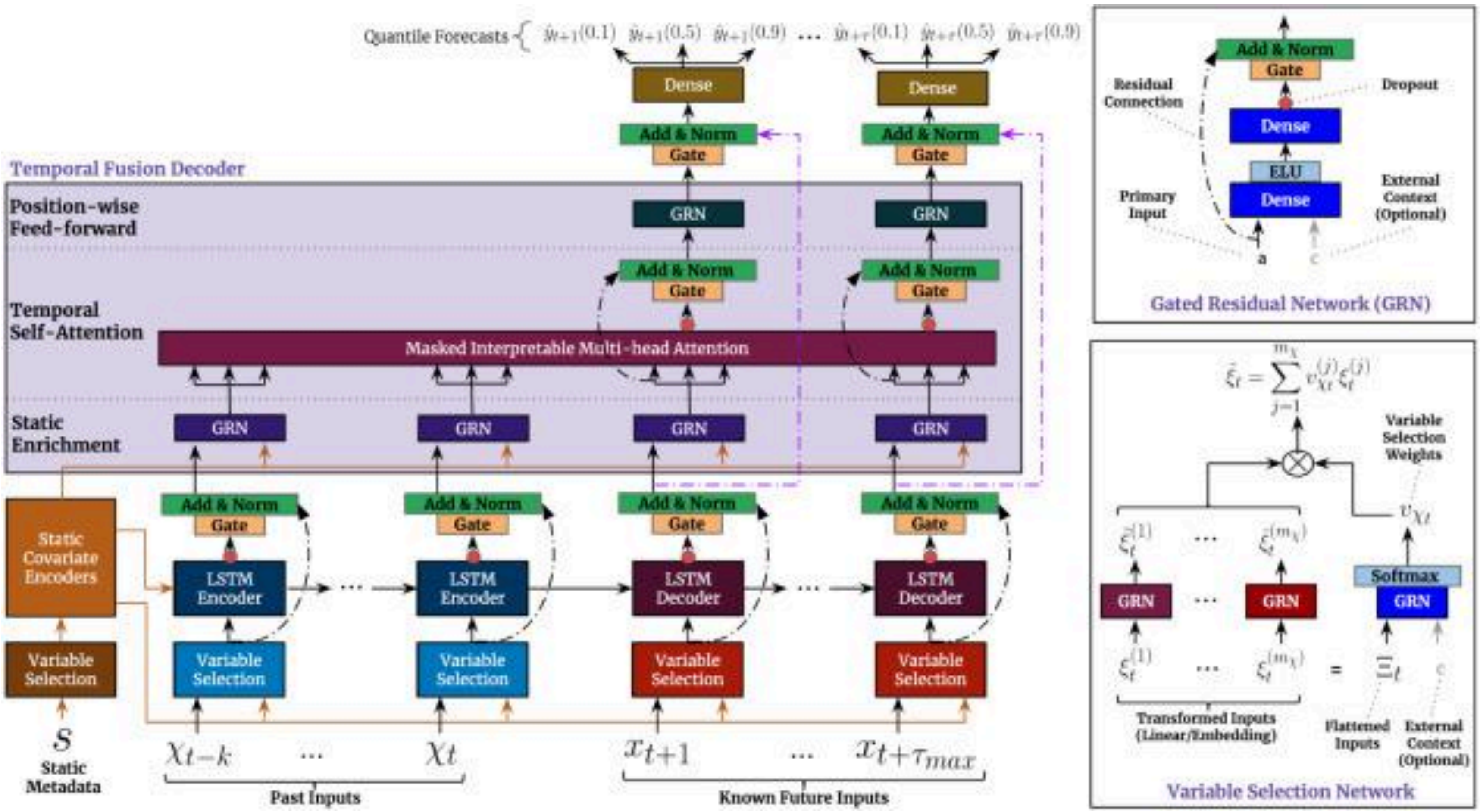
- Example: Temporal Fusion Transformers (Lim et al., 2021)



https://doi.org/10.1016/j.ijforecast.2021.03.012

Attention is all you need (Transformers)

- Example: Temporal Fusion Transformers (Lim et al., 2021)



https://doi.org/10.1016/j.ijforecast.2021.03.012

Attention is all you need (Transformers)

- Example: Temporal Fusion Transformers (Lim et al., 2021)

(a) S&P 500

(b) FTSE 100

(c) Nikkei 225

Fig. E.7. TFT one-step-ahead forecasts for realized volatility across major stock indices following the incidence of COVID-19. From the target values in purple, we note the regime shift following the sharp increase in index volatilities in March 2020, which later decay to elevated baseline levels over 2020. While volatilities in March 2020 do appear at or around the 0.975 quantile forecast, the TFT maintains reasonable uncertainty estimates over the regime shift – with most values lying within the 95% credible interval – demonstrating its ability to adapt to changing temporal dynamics. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

https://doi.org/10.1016/j.ijforecast.2021.03.012

Attention is all you need (Transformers)

- Example: Informer (Zhou et al., 2021)

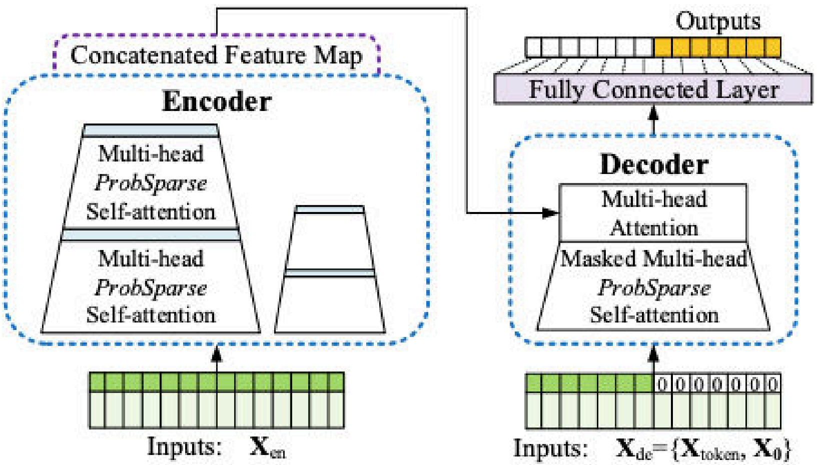


Figure 2: Informer model overview. Left: The encoder receives massive long sequence inputs (green series). We replace canonical self-attention with the proposed *ProbSparse* self-attention. The blue trapezoid is the self-attention distilling operation to extract dominating attention, reducing the network size sharply. The layer stacking replicas increase robustness. Right: The decoder receives long sequence inputs, pads the target elements into zero, measures the weighted attention composition of the feature map, and instantly predicts output elements (orange series) in a generative style.

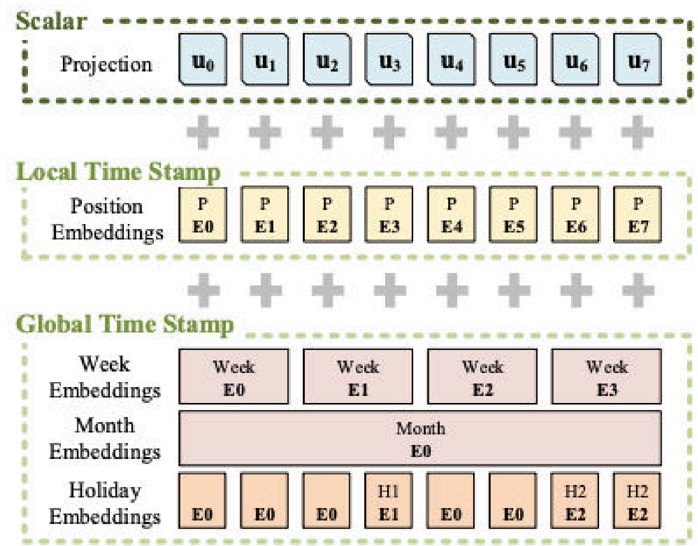


Figure 6: The input representation of Informer. The inputs's embedding consists of three separate parts, a scalar projection, the local time stamp (Position) and global time stamp embeddings (Minutes, Hours, Week, Month, Holiday etc.).

https://doi.org/10.1609/aaai.v35i12.17325

Attention is all you need (Transformers)

- Example: Autoformer (Wu et al., 2021)

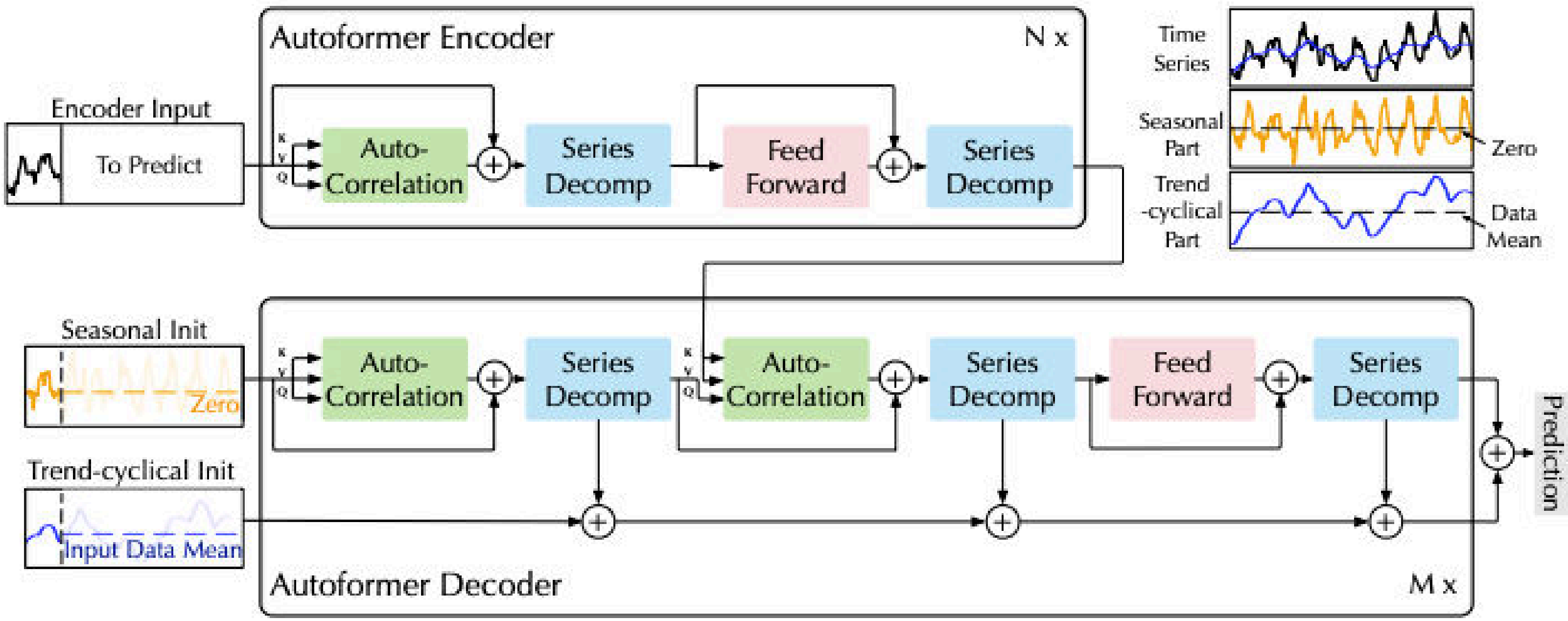


Figure 1: Autoformer architecture. The encoder eliminates the long-term trend-cyclical part by series decomposition blocks (blue blocks) and focuses on seasonal patterns modeling. The decoder accumulates the trend part extracted from hidden variables progressively. The past seasonal information from encoder is utilized by the encoder-decoder Auto-Correlation (center green block in decoder).

https://proceedings.neurips.cc/paper/2021/hash/bcc0d400288793e8bdcd7c19a8ac0c2b-Abstract.html

Attention is all you need (Transformers)

- Example: Autoformer (Wu et al., 2021)

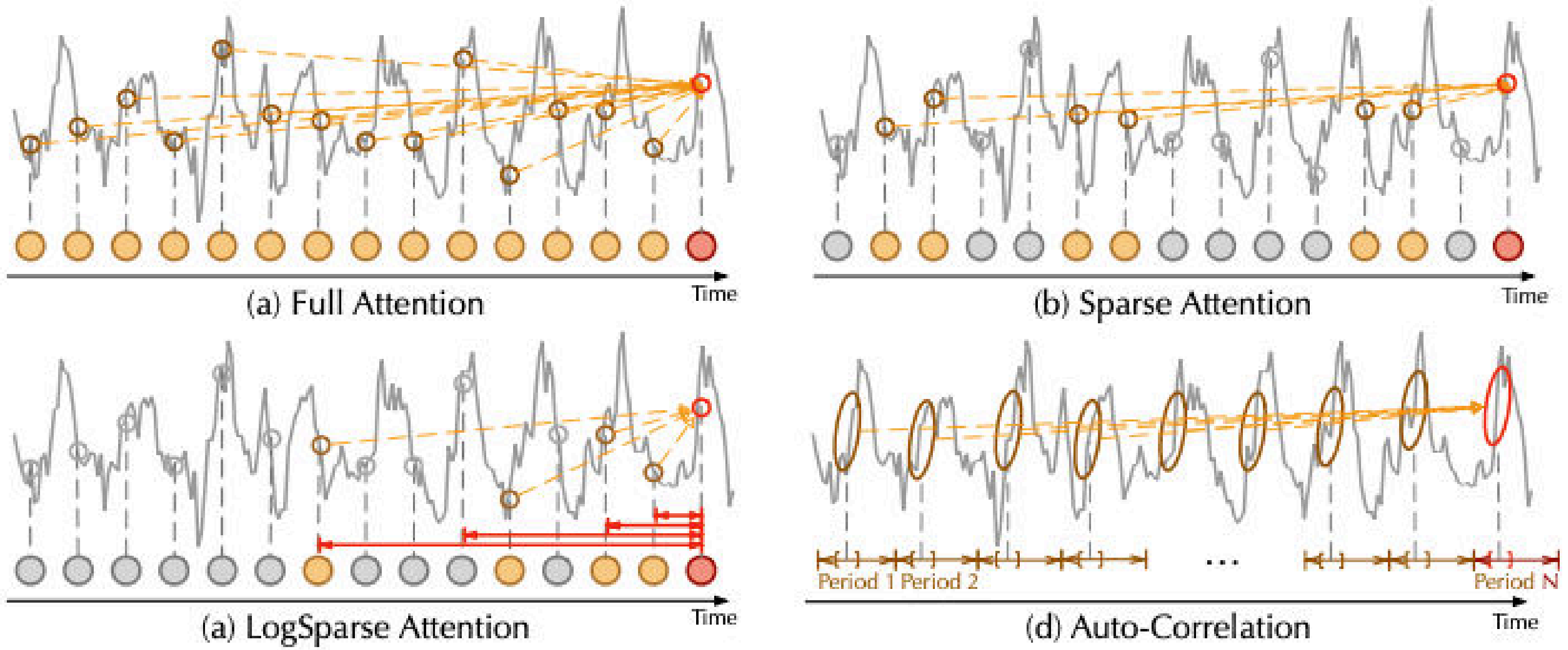


Figure 3: Auto-Correlation vs. self-attention family. Full Attention [41] (a) adapts the fully connection among all time points. Sparse Attention [23, 48] (b) selects points based on the proposed similarity metrics. LogSparse Attention [26] (c) chooses points following the exponentially increasing intervals. Auto-Correlation (d) focuses on the connections of sub-series among underlying periods.

https://proceedings.neurips.cc/paper/2021/hash/bcc0d400288793e8bdcd7c19a8ac0c2b-Abstract.html

2.4

LLM-based models

LLM-based models

- LLMTime[1]

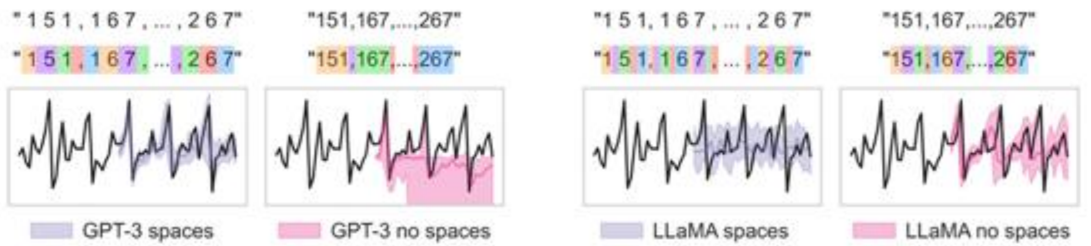


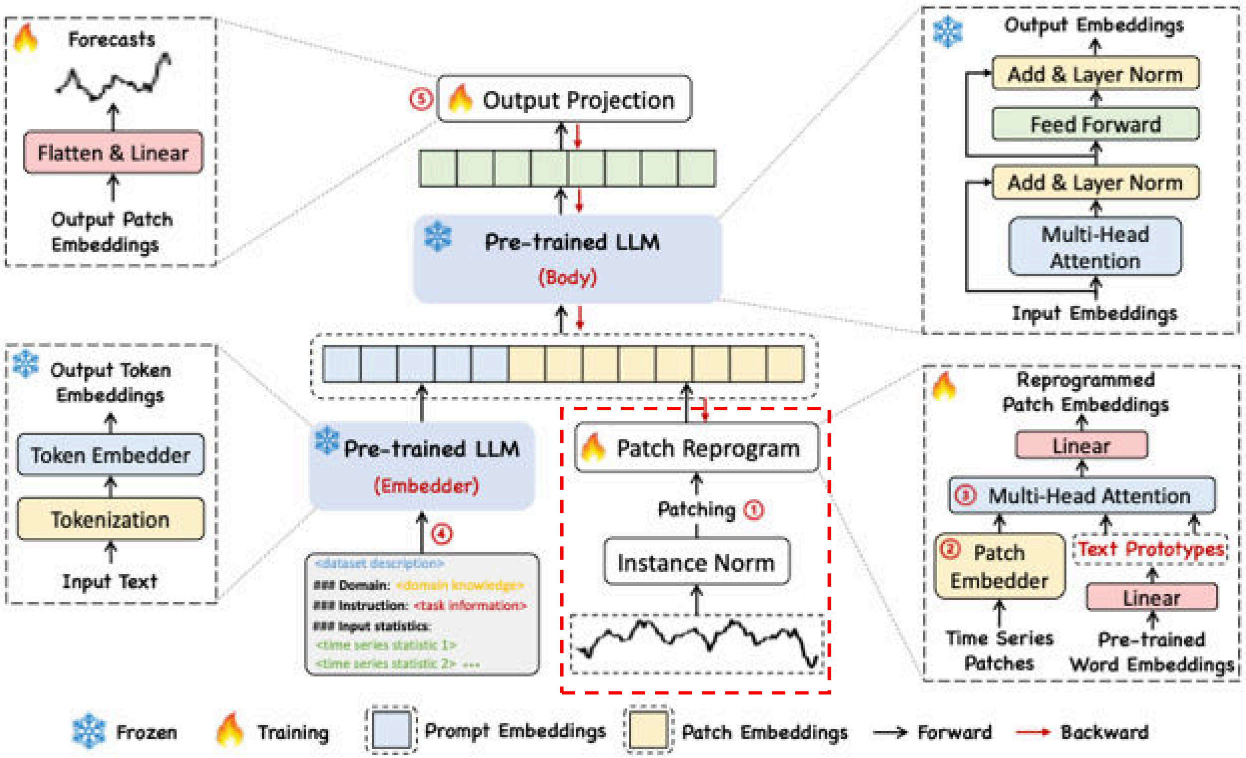
Figure 2: Careful tokenization is important for good forecasting with LLMs. Using the Australian Wine dataset from Darts [24], with values [151, 167, ..., 267], we show the tokenization used by GPT-3 [9] and LLaMA-2 [45] and the corresponding effect on forecasting performance. Added spaces allow GPT-3 to create one token per digit, leading to good performance. LLaMA-2, on the other hand, tokenizes digits individually, and adding spaces hurts performance.

  - Tokenization for the numeric values
  - (-): 연산/대수관계를 제대로 이해하는 방식이 아니라, 한계가 명확함

[1] Gruver, N., Finzi, M., Qiu, S., & Wilson, A. G. (2023). Large language models are zero-shot time series forecasters. *Advances in Neural Information Processing Systems*, *36*, 19622-19635.

LLM-based models

- TimeLLM [2]

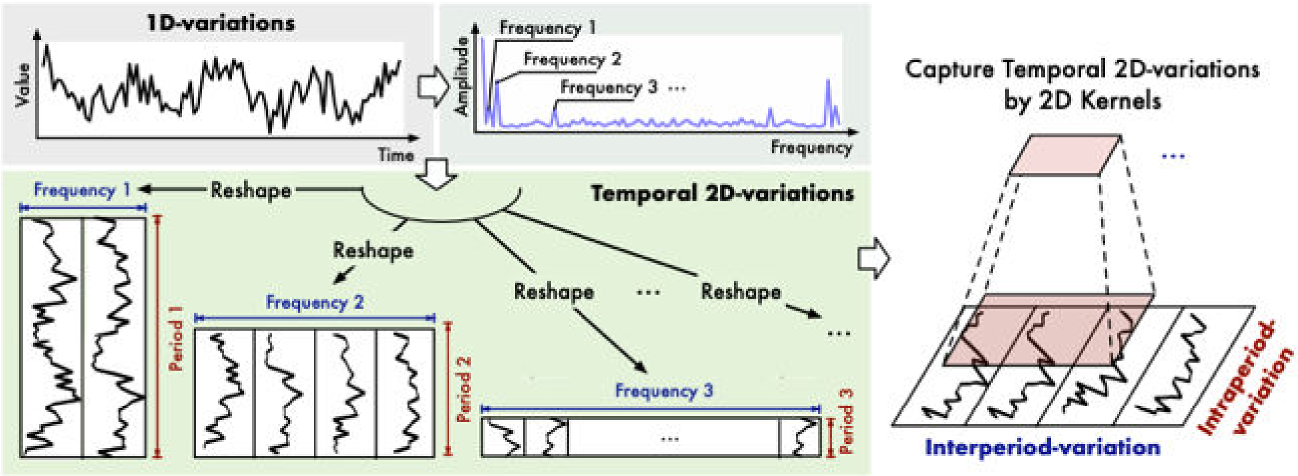


– Projecting TS to single-modality embedding
– (-): Foundation language model을 backbone으로 사용하기에, modality gap/ TS -> emb 로 부터 발생하는 정보 손실

[1] Jin, M., Wang, S., Ma, L., Chu, Z., Zhang, J. Y., Shi, X., ... & Wen, Q. (2023). Time-llm: Time series forecasting by reprogramming large language models. *arXiv preprint arXiv:2310.01728.*

LLM-based models

- TimesNet [3]



  - TS -> frequency / amplitude로의 변환을 통해 long / short context 정보를 동시에 반영하려 함.
  - (-): domain shift 발생 시 모델이 가지고 있는 inductive bias가 상당 부분 사라짐 -> adaptation에 어려움을 겪을 가능성 높음.

[3] Wu, H., Hu, T., Liu, Y., Zhou, H., Wang, J., & Long, M. (2022). Timesnet: Temporal 2d-variation modeling for general time series analysis. *arXiv preprint arXiv:2210.02186.*

LLM-based models

- Time-VLM

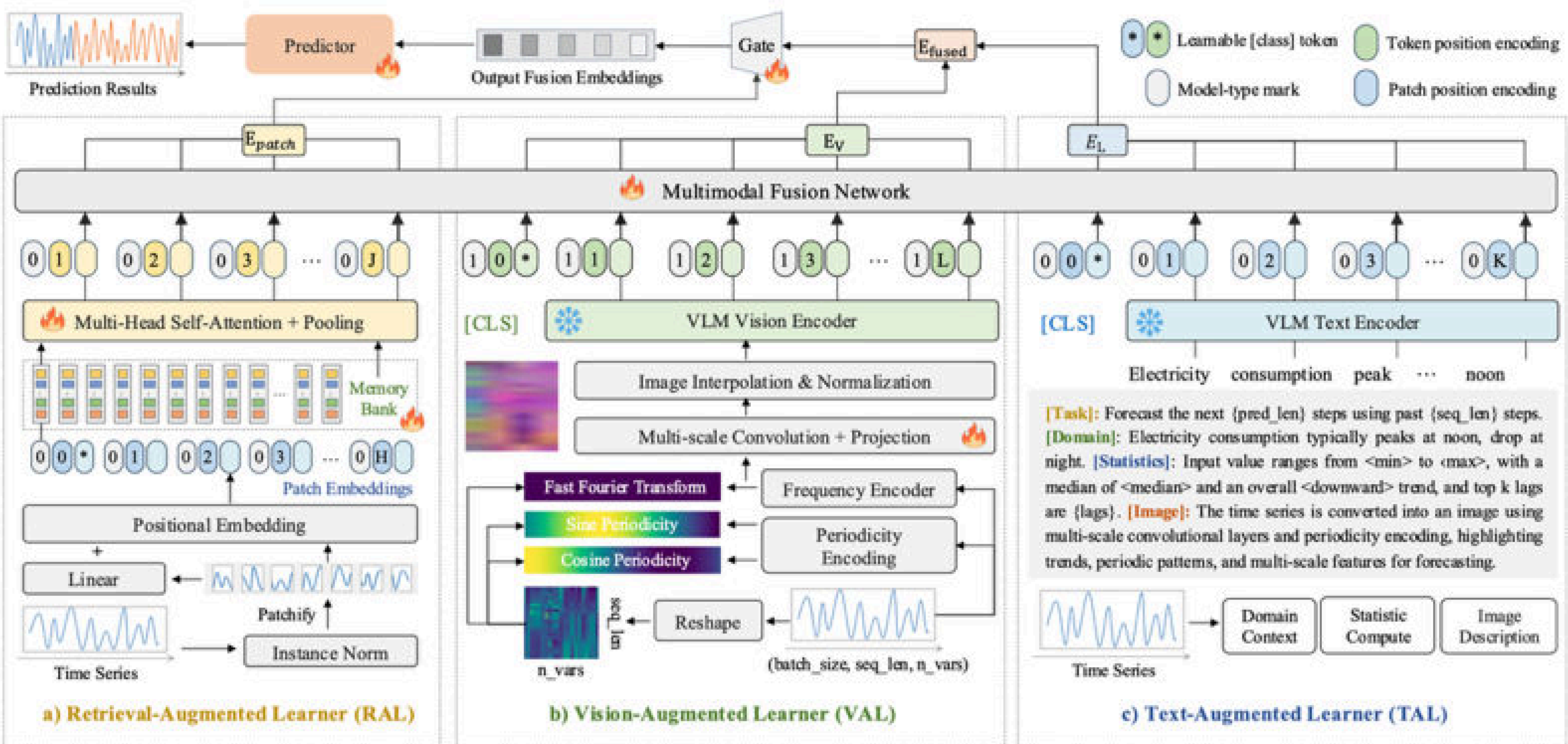


Figure 2: Overview of the Time-VLM framework.

Zhong, S., Ruan, W., Jin, M., Li, H., Wen, Q., & Liang, Y. (2025). Time-VLM: Exploring multimodal vision-language models for augmented time series forecasting. *arXiv preprint arXiv:2502.04395*.

LLM-based models

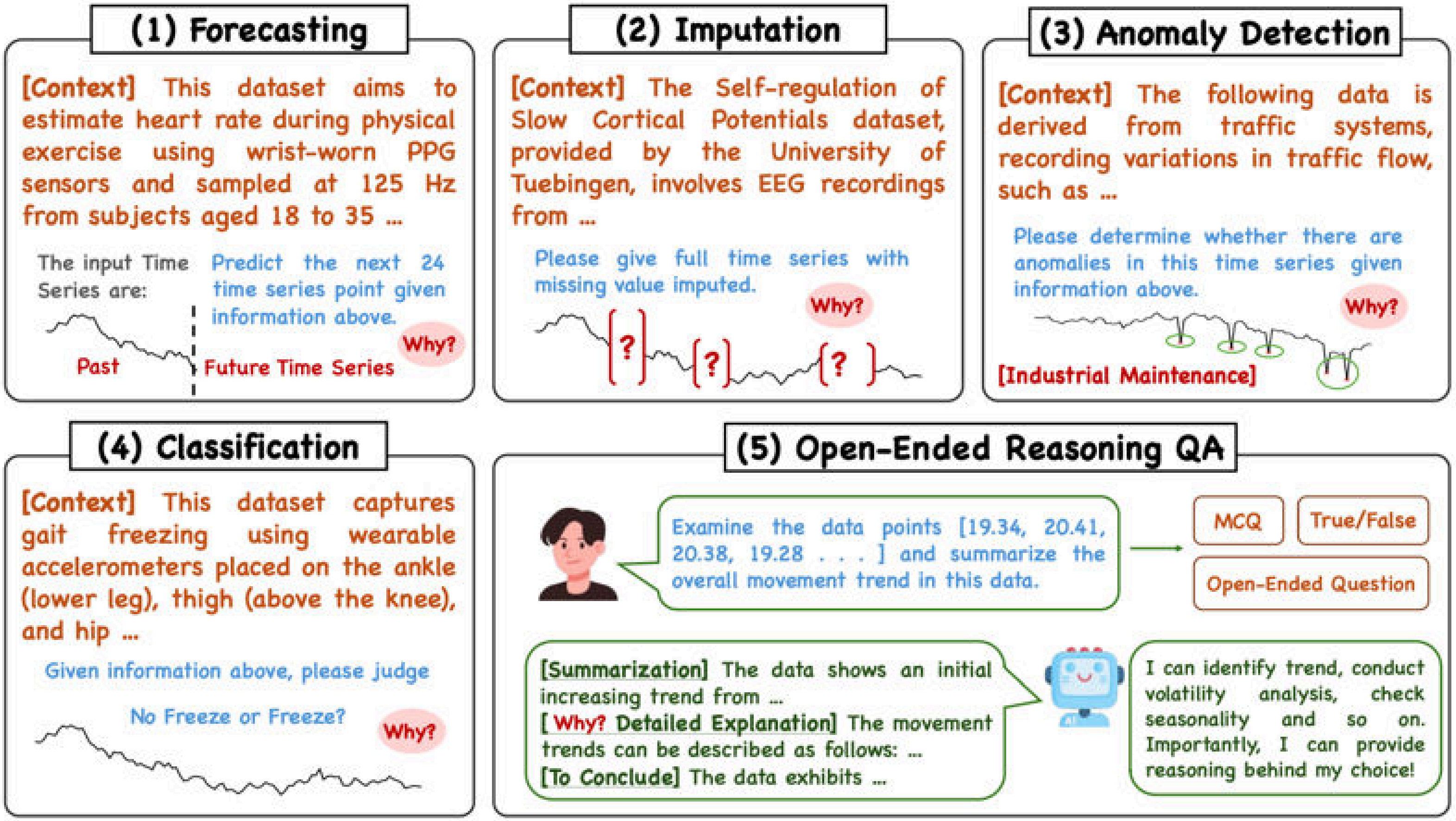


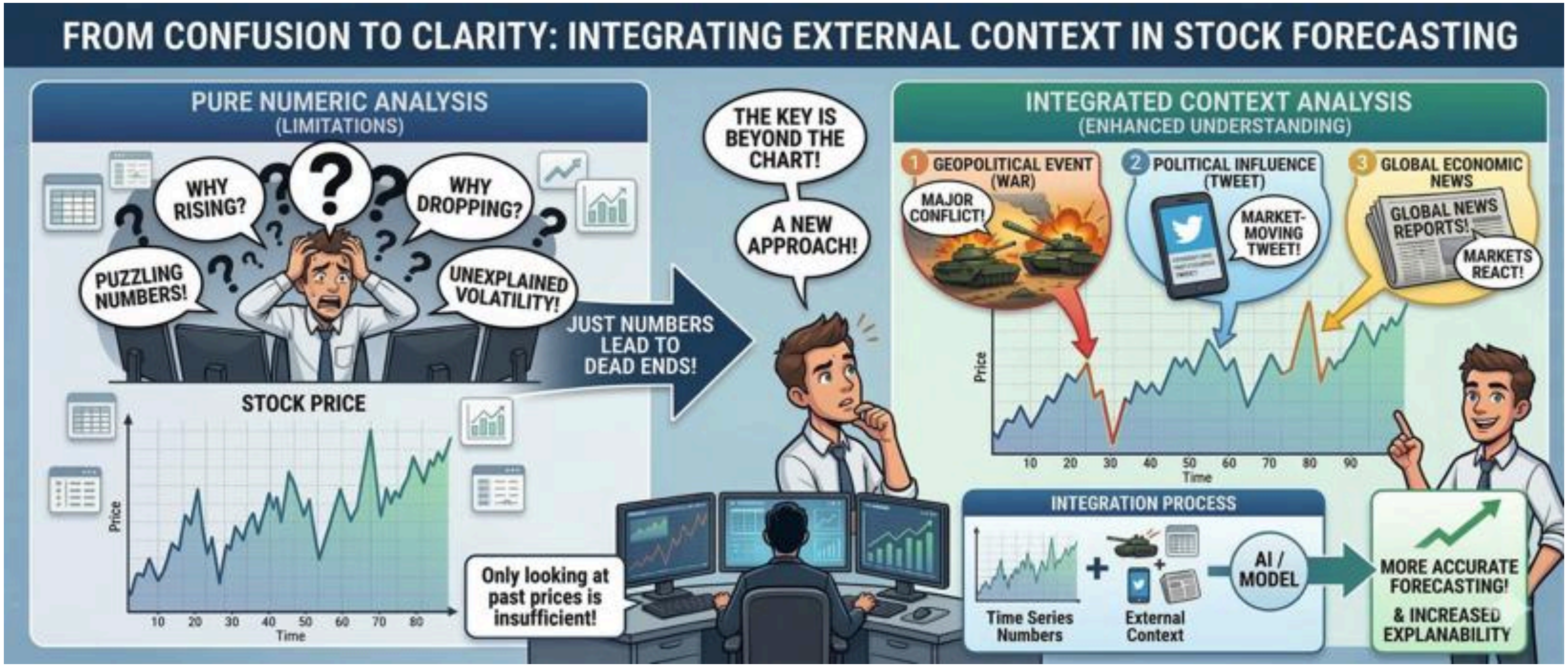
Figure 3: The demonstration of the Time-MQA with context enhancement.

Kong, Y., Yang, Y., Hwang, Y., Du, W., Zohren, S., Wang, Z., ... & Wen, Q. (2025). Time-mqa: Time series multi-task question answering wit h context enhancement. arXiv preprint arXiv:2503.01875.

Section 3

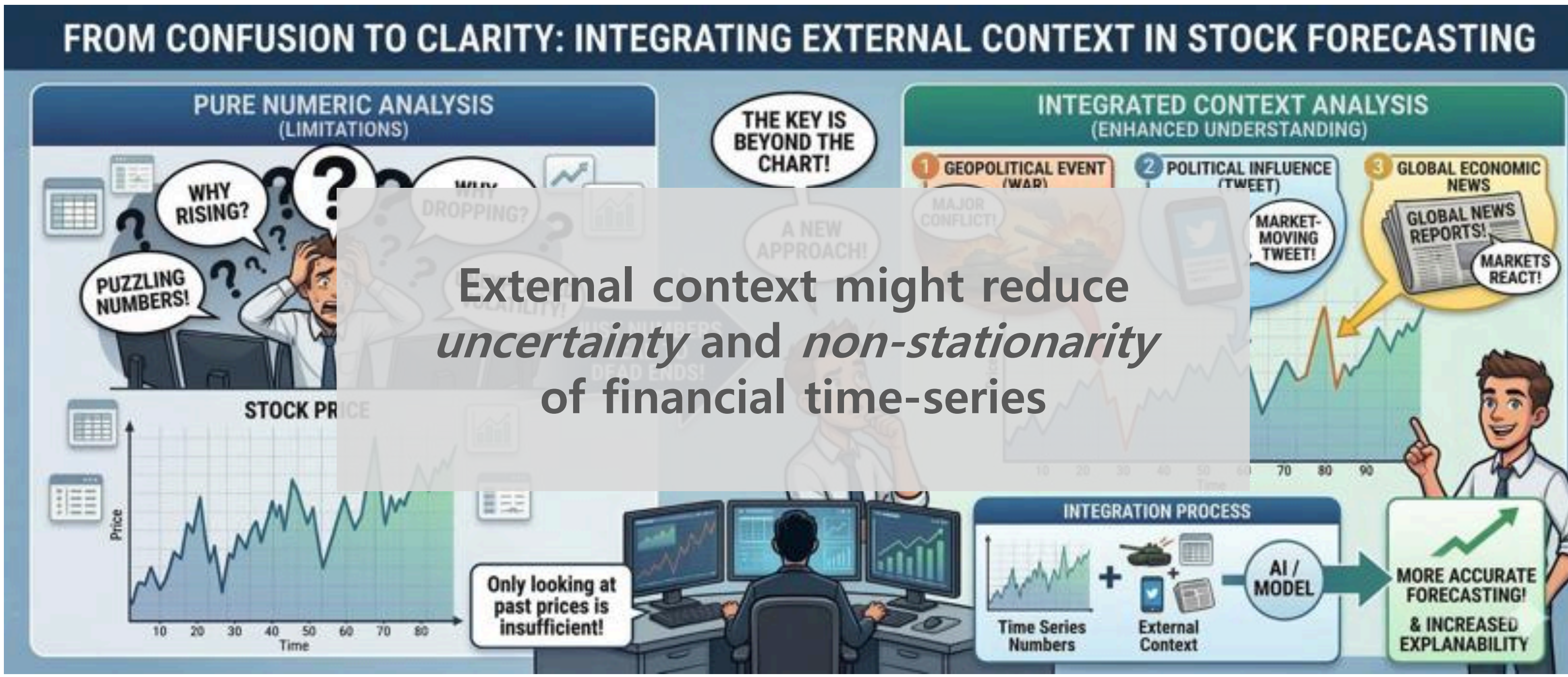
Case Studies

Motivation



65

Motivation



External context might reduce *uncertainty* and *non-stationarity* of financial time-series

66

Paper 1

Return Prediction for Mean-Variance Portfolio Selection: How Decision-Focused Learning Shapes Forecasting Models

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Abstract

Markowitz laid the foundation of portfolio theory through mean-variance optimization (MVO). However, MVO's effectiveness depends on precise estimation of expected returns, variances, and covariances, which are typically uncertain. Machine learning models are increasingly used to estimate these parameters, trained to minimize prediction errors like MSE, which treats errors uniformly across assets. Recent studies show this leads to suboptimal decisions and propose Decision-Focused Learning (DFL), integrating prediction and optimization to improve outcomes. While studies demonstrate DFL's potential to enhance portfolio performance, the mechanisms of how DFL modifies prediction models for MVO remain unexplored. This study investigates how DFL adjusts stock return prediction models to optimize MVO decisions. We show that DFL's gradient tilts MSE-based prediction errors by the inverse covariance matrix Σ^{-1}, incorporating inter-asset correlations into learning, while MSE treats each asset independently. This tilting creates systematic biases where DFL overestimates returns for portfolio assets while underestimating excluded assets. Our findings reveal why DFL achieves superior performance despite higher prediction errors. The strategic biases are features, not flaws.

1 Introduction

Decision-making is a crucial aspect across various fields, where optimization is often employed to guide us toward the best possible outcome quantitatively. In ideal scenarios, where the values of all parameters in the optimization formula are known, a mathematically optimal solution can be determined with precision. However, in most real-world situations, parameters are often uncertain. The quality of the decisions derived from optimization heavily depends on how accurately these uncertain parameters are estimated. Therefore, it is essential to ensure accurate estimation of these parameters to achieve high-quality decision-making outcomes.

This is particularly evident in the field of portfolio optimization, where financial decisions are made under uncertainty. Harry Markowitz laid the foundation of portfolio theory based on mean-variance optimization (MVO) [24]. The objective of MVO is to construct an investment portfolio that maximizes return for a given level of risk or minimizes risk for a given level of expected return. While the MVO has been fundamental to investment management [19], its effectiveness depends on the accurate estimates of the expected return, variance, and covariance of asset returns, which are often uncertain in practice. Despite this uncertainty, decision makers who utilize MVO should have their own estimates of expected

Lee, J., Jeon, H., Bae, H., & Lee, Y. (2025, November). Return Prediction for Mean-Variance Portfolio Selection: How Decision-Focused Learning Shapes Forecasting Models. In Proceedings of the 6th ACM International Conference on AI in Finance (pp. 114-122). https://dl.acm.org/doi/full/10.1145/3768292.3770423

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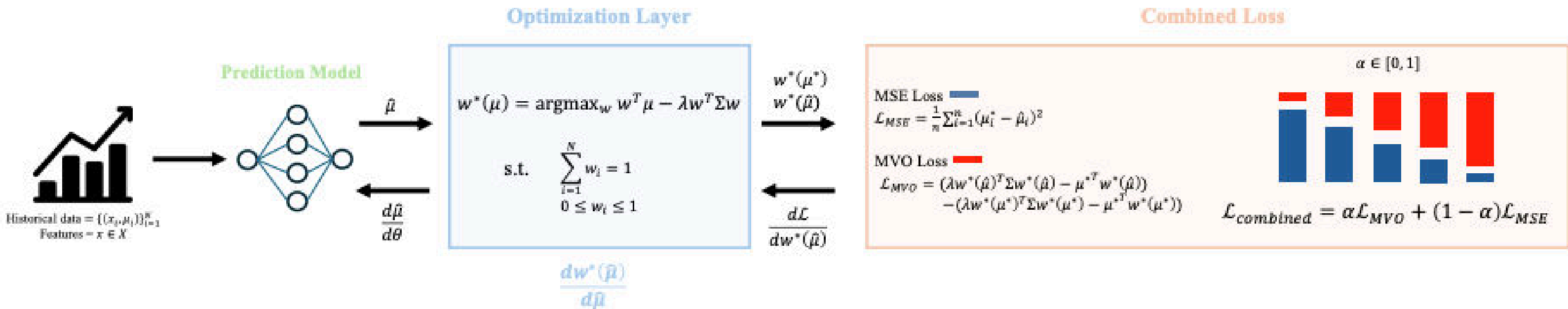

Figure 1: DFL training procedure for MVO. The prediction model outputs $\hat{\mu}$, which determines portfolio weights $w^*(\hat{\mu})$ through the optimization layer. The combined loss $\mathcal{L}_{combined} = \alpha\mathcal{L}_{MVO} + (1-\alpha)\mathcal{L}_{MSE}$ balances prediction accuracy and decision quality. The optimization layer enables computation of $\frac{dw^*(\hat{\mu})}{d\hat{\mu}}$, which is necessary for calculating the MVO loss gradient.

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$$\mathcal{L}_{\text{Combined}} = \alpha \mathcal{L}_{\text{MVO}} + (1 - \alpha)\mathcal{L}_{\text{MSE}} \quad (12)$$

Table 2: Portfolio performance metrics for varying λ and α. Models with $\alpha > 0$ consistently outperform pure MSE ($\alpha = 0$), with optimal performance typically at intermediate values ($\alpha \in [0.25, 0.75]$). DFL improves both returns and risk metrics, achieving higher Sharpe ratios and often lower maximum drawdowns. Bold values indicate best performance for each metrics.

Panel A. DOW 30 Dataset						Panel B. S&P 100 Dataset					
λ	α	Return (↑)	Sharpe (↑)	MDD (↓)	Wealth (↑)	λ	α	Return (↑)	Sharpe (↑)	MDD (↓)	Wealth (↑)
	0.00	0.181	0.681	0.253	1.459		0.00	0.209	0.734	0.266	1.184
	0.25	0.323	1.138	0.230	1.669		0.25	0.256	0.744	0.327	1.673
0.1	0.50	**0.398**	**1.228**	0.261	**2.138**	0.1	0.50	0.134	0.513	0.278	1.239
	0.75	0.177	0.699	0.268	1.408		0.75	0.091	0.386	0.319	1.277
	1.00	0.143	0.593	**0.226**	1.275		1.00	**0.292**	**1.179**	**0.249**	**1.701**
	0.00	0.125	0.529	0.251	1.250		0.00	0.153	0.578	0.276	1.126
	0.25	0.192	0.754	0.258	1.459		0.25	0.187	0.769	0.295	1.339
0.5	0.50	**0.347**	**1.302**	**0.211**	**1.813**	0.5	0.50	0.231	0.701	0.295	1.680
	0.75	0.248	1.051	0.230	1.520		0.75	**0.303**	**1.217**	**0.230**	**1.763**
	1.00	0.151	0.641	0.268	1.354		1.00	0.183	0.769	0.267	1.512
	0.00	0.115	0.519	0.242	1.234		0.00	0.129	0.540	0.242	1.090
	0.25	0.165	0.775	0.221	1.383		0.25	0.237	0.746	0.267	1.772
1.0	0.50	**0.327**	1.150	0.218	**1.669**	1.0	0.50	**0.319**	**1.260**	**0.214**	**1.773**
	0.75	0.312	**1.311**	**0.203**	1.651		0.75	0.201	0.801	0.237	1.410
	1.00	0.180	0.763	0.256	1.397		1.00	0.252	0.949	0.287	1.569
	0.00	0.091	0.531	**0.195**	1.163		0.00	0.144	0.838	**0.194**	1.258
	0.25	**0.217**	**0.932**	0.203	**1.566**		0.25	0.203	0.831	0.250	1.523
5.0	0.50	0.187	0.756	0.215	1.473	5.0	0.50	0.136	0.585	0.219	1.347
	0.75	0.197	0.871	0.211	1.379		0.75	0.193	0.855	0.237	1.464
	1.00	0.164	0.738	0.223	1.361		1.00	**0.213**	**0.928**	0.276	**1.611**

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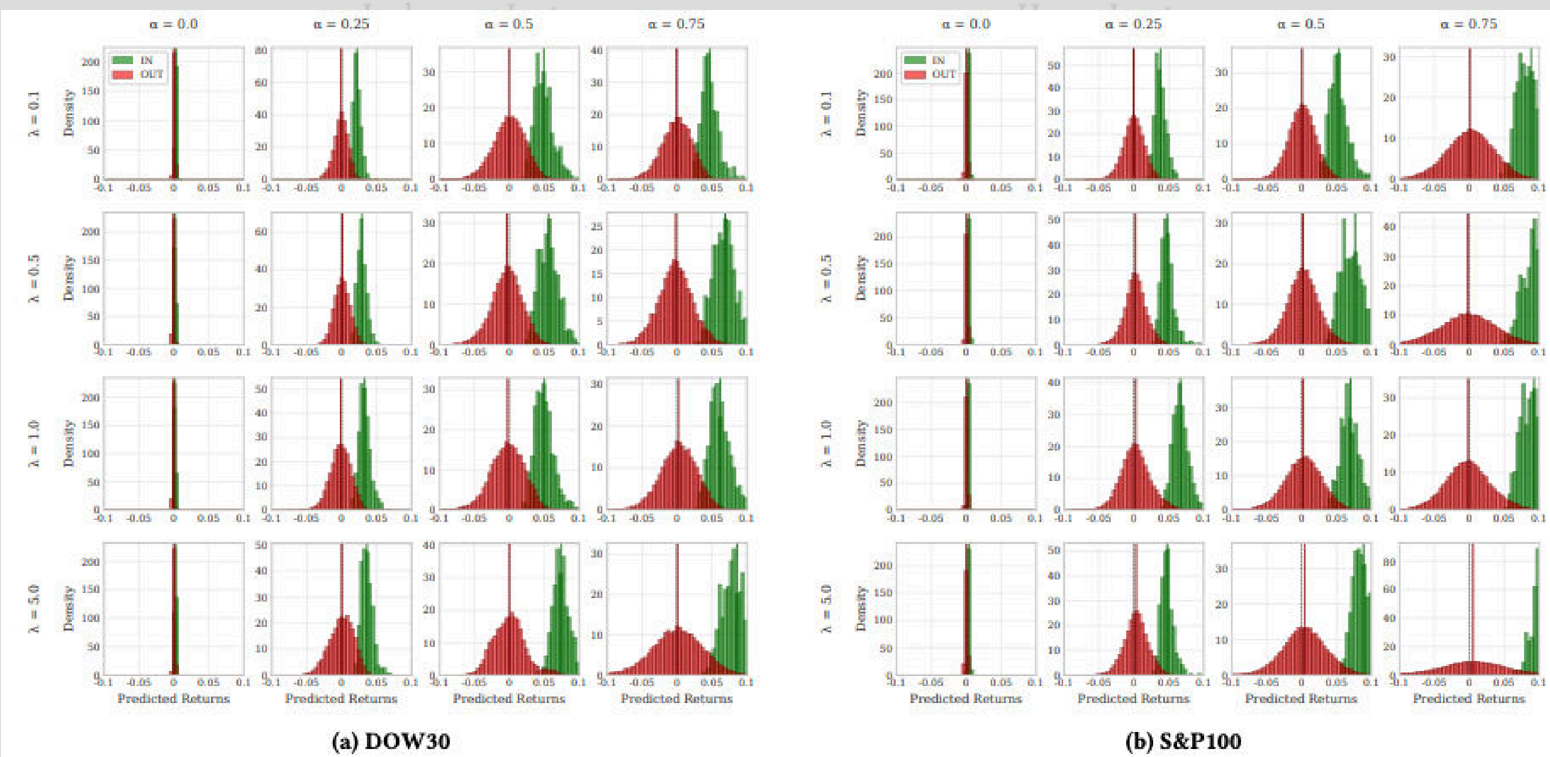

Figure 4: Predicted return distributions for IN/OUT portfolio groups across different λ and α values. As α increases, the separation between IN and OUT group distributions widens. The case of $\alpha = 1$ is excluded due to extreme distribution separation.

ings reveal why DFL achieves superior performance despite higher prediction errors. The strategic biases are features, not flaws.

pected return, variance, and covariance of asset returns, which are often uncertain in practice. Despite this uncertainty, decision makers who utilize MVO should have their own estimates of expected

Lee, J., Jeon, H., Bae, H., & Lee, Y. (2025, November). Return Prediction for Mean-Variance Portfolio Selection: How Decision-Focused Learning Shapes Forecasting Models. In Proceedings of the 6th ACM International Conference on AI in Finance (pp. 114-122). https://dl.acm.org/doi/full/10.1145/3768292.3770423

Paper 2

Estimating Covariance for Global Minimum Variance Portfolio: A Decision-Focused Learning Approach

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Abstract

Portfolio optimization constitutes a cornerstone of risk management by quantifying the risk-return trade-off. Since it inherently depends on accurate parameter estimation under conditions of future uncertainty, the selection of appropriate input parameters is critical for effective portfolio construction. However, most conventional statistical estimators and machine learning algorithms determine these parameters by minimizing mean-squared error (MSE), a criterion that can yield suboptimal investment decisions.

In this paper, we adopt decision-focused learning (DFL)—an approach that directly optimizes decision quality rather than prediction error such as MSE—to derive the global minimum-variance portfolio (GMVP). Specifically, we theoretically derive the gradient of decision loss using the analytic solution of GMVP and its properties regarding the principal components of itself. Through extensive empirical evaluation, we show that prediction-focused estimation methods may fail to produce optimal allocations in practice, whereas DFL-based methods consistently deliver superior decision performance. Furthermore, we provide a comprehensive analysis of DFL's mechanism in GMVP construction, focusing on its volatility reduction capability, decision-driving features, and estimation characteristics.

1 Introduction

Markowitz's seminal work introduced mean–variance optimization (MVO) [29], which provides a quantitative framework for balancing risk and return and has since become modern portfolio theory's foundation. The most important part in MVO is parameter estimation, due to its high sensitivity [21]. Practically, MVO parameters are typically calibrated using an in-sample period under the assumption of return stationarity, which results in poor performances because the distribution underlying the in-sample and out-of-sample periods often differs. Empirical studies have shown that MVO portfolios constructed with estimated parameters can underperform naïve diversification strategies like the equally weighted portfolio, highlighting appropriate parameter estimation [15].

Global minimum-variance portfolio (GMVP) constitutes the least risky portfolio in the efficient frontier, therefore it inherits a key for understanding the modern portfolio theory. Since GMVP depends solely on the covariance matrix, not on expected returns, its parameter uncertainty becomes much lower than in the MVO framework. Moreover, empirical evidence indicates that GMVPs often achieve superior out-of-sample return performance compared to market-cap-weighted benchmarks [12, 13] and exhibit lower realized volatility [6]. Because GMVP construction requires estimating only covariances—unlike MVO, which requires both ex-

Kim, J., Tae, I., & Lee, Y. (2025, November). Estimating Covariance for Global Minimum Variance Portfolio: A Decision-Focused Learning Approach. In Proceedings of the 6th ACM International Conference on AI in Finance (pp. 105-113).
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Paper 2

Estimating Covariance for Global Minimum Variance Portfolio: A Decision-Focused Learning Approach

Juchan Kim* Inwoo Tae* Yongjae Lee†

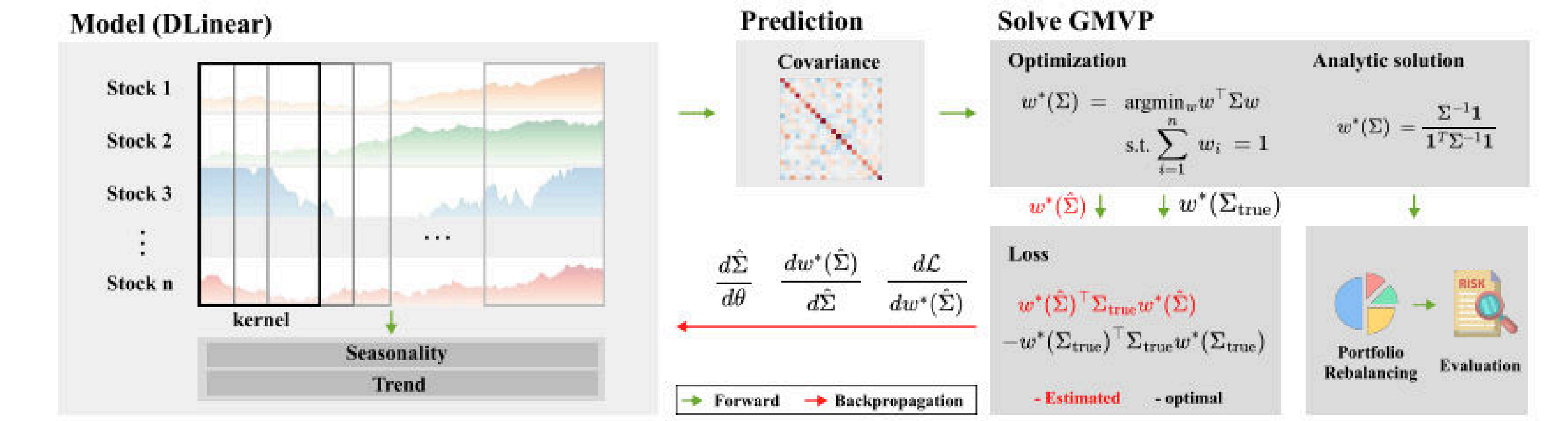

Figure 1: DFL framework for GMVP construction. Historical returns of N assets are processed by DLinear to predict covariance matrix Σ. The GMVP optimization layer computes portfolio weights w^* using the closed form solution, and the regret loss is applied to minimize the gap between predicted and optimal portfolio volatility for end-to-end training.

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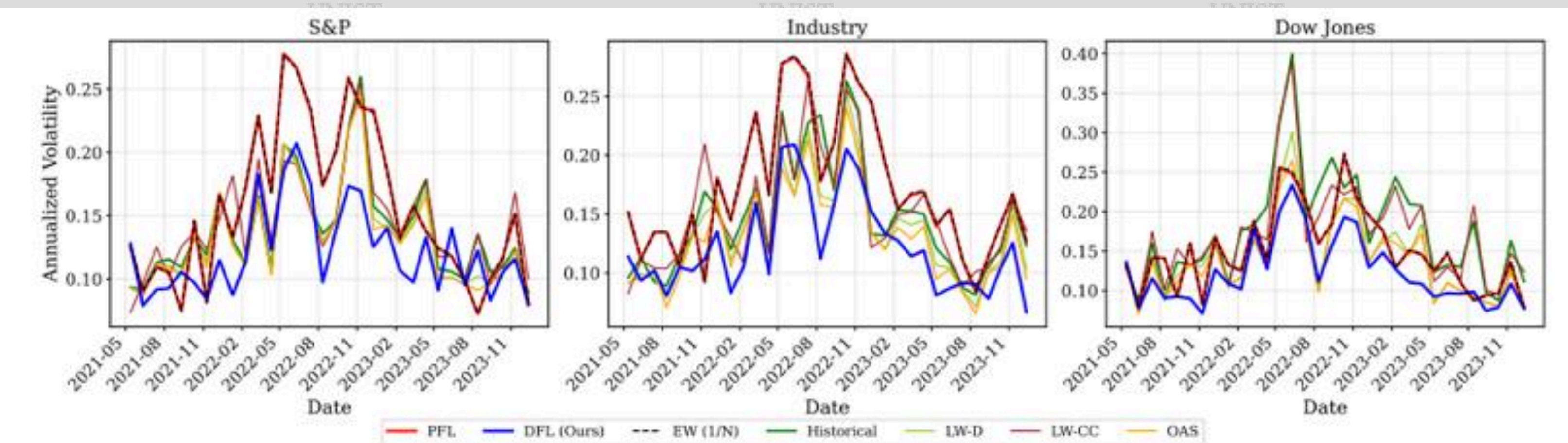

Figure 3: Annualized volatility of each model's portfolio with $\delta_{in} = \delta_{out} = 21$ from the left to the right, respectively.

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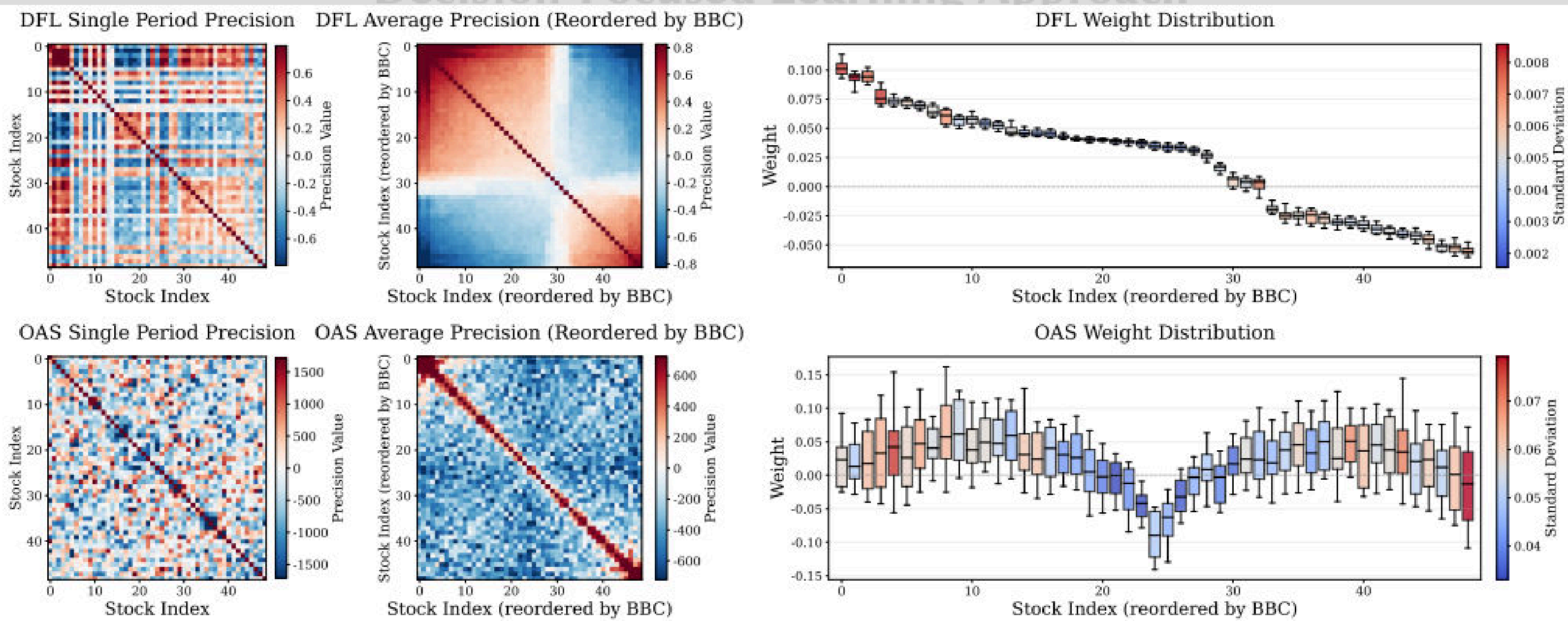

Figure 6: Precision matrix derived by DFL and OAS for single test period in the Industry dataset (left), Average of the permuted precision matrix estimated by DFL and OAS for the entire test period in the Industry dataset (middle), Average and standard deviation of GMVP weights in the reordered index for the entire test period in the Industry dataset (right).

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Paper 3 (working paper)

- Applied decision-focused learning to MVO with LLM-based prediction

Decision-informed Neural Networks with Large Language Model Integration for Portfolio Optimization

Yoontae Hwang[1], Yaxuan Kong[1], Stefan Zohren[1], Yongjae Lee[2]

[1]University of Oxford [2]Ulsan National Institute of Science and Technology(UNIST)

(Jan 31, 2025)

This paper addresses the critical disconnect between prediction and decision quality in portfolio optimization by integrating Large Language Models (LLMs) with decision-focused learning. We demonstrate both theoretically and empirically that minimizing the prediction error alone leads to suboptimal portfolio decisions. We aim to exploit the representational power of LLMs for investment decisions. An attention mechanism processes asset relationships, temporal dependencies, and macro variables, which are then directly integrated into a portfolio optimization layer. This enables the model to capture complex market dynamics and align predictions with the decision objectives. Extensive experiments on S&P100 and DOW30 datasets show that our model consistently outperforms state-of-the-art deep learning models. In addition, gradient-based analyses show that our model prioritizes the assets most crucial to decision making, thus mitigating the effects of prediction errors on portfolio performance. These findings underscore the value of integrating decision objectives into predictions for more robust and context-aware portfolio management.

Keywords: Portfolio Optimization, Large Language Models, Decision-Focused Learning, Estimation Error

JEL Classification: G11, G17, G45, G61, C53

1. Introduction

The estimation of parameters for portfolio optimization has long been recognized as one of the most challenging aspects of implementing modern portfolio theory (Michaud 1989, DeMiguel *et al.* 2009). While Markowitz's mean-variance framework (Markowitz 1952) provides an elegant theoretical foundation for portfolio selection, its practical implementation has been persistently undermined by estimation errors in input parameters. (Chopra and Ziemba 1993, Chung *et al.* 2022) demonstrate that small changes in estimated expected returns can lead to dramatic shifts in optimal, while (Ledoit and Wolf 2003, 2004) show that traditional sample covariance estimates become unreliable as the number of assets grows relative to the sample size.

These estimation challenges are exacerbated by a fundamental limitation in the conventional approach to portfolio optimization: the reliance on a sequential, two-stage process where parameters are first estimated from historical data, and these estimates are then used as inputs in the optimization problem. This methodology, while computationally convenient, creates a profound disconnect between prediction accuracy and decision quality. Recent evidence suggests that even substantial

Hwang, Y., Kong, Y., Zohren, S., & Lee, Y. (2025). Decision-informed neural networks with large language model integration for portfolio optimization. arXiv preprint arXiv:2502.00828.
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Paper 3 (working paper)

- Applied D
  prediction

Decision-informed Neural Networks with Large Language Model Integration for Portfolio Optimization

Yoontae Hwang[1], Yaxuan Kong[1], Stefan Zohren[1], Yongjae Lee[2]

[1]University of Oxford [2]Ulsan National Institute of Science and Technology(UNIST)

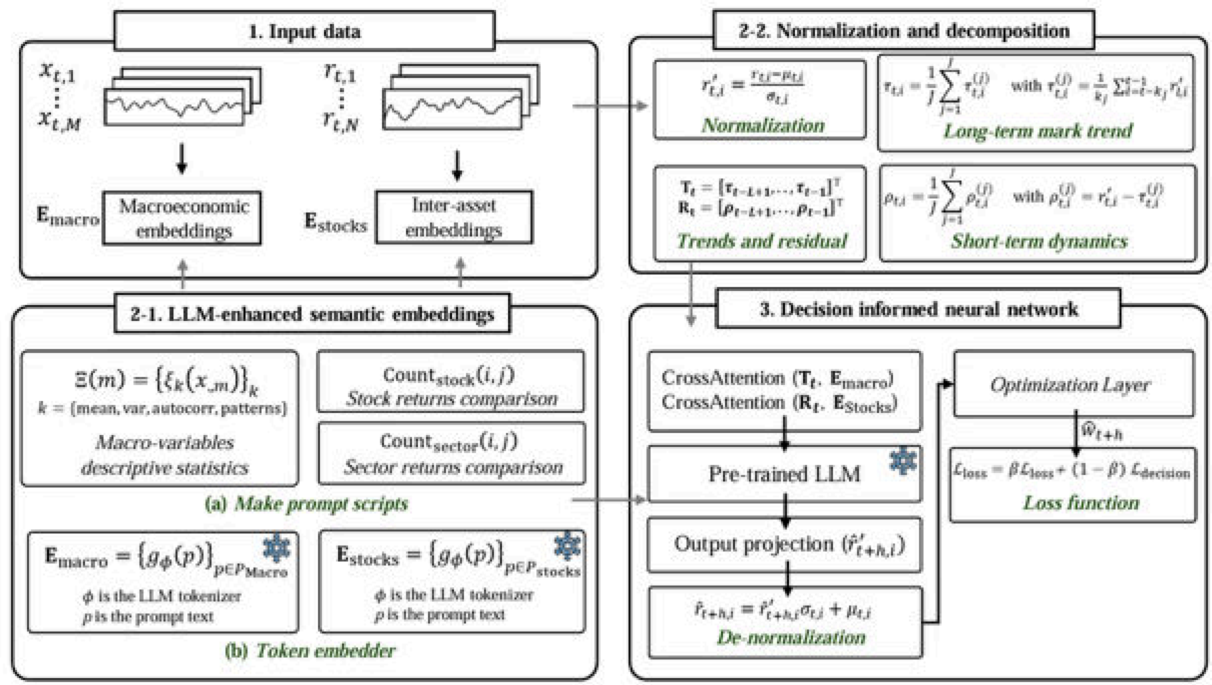

Figure 1.Schematic of the proposed Decision-Informed Neural Network (DINN) architecture for unified return forecasting and portfolio selection. The entire system is trained end-to-end to align predictive accuracy with decision quality.

Hwang, Y., Kong, Y., Zohren, S., & Lee, Y. (2025). Decision-informed neural networks with larg
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Paper 3 (working paper)

Decision-informed Neural Networks with Large Language Model Integration for Portfolio Optimization

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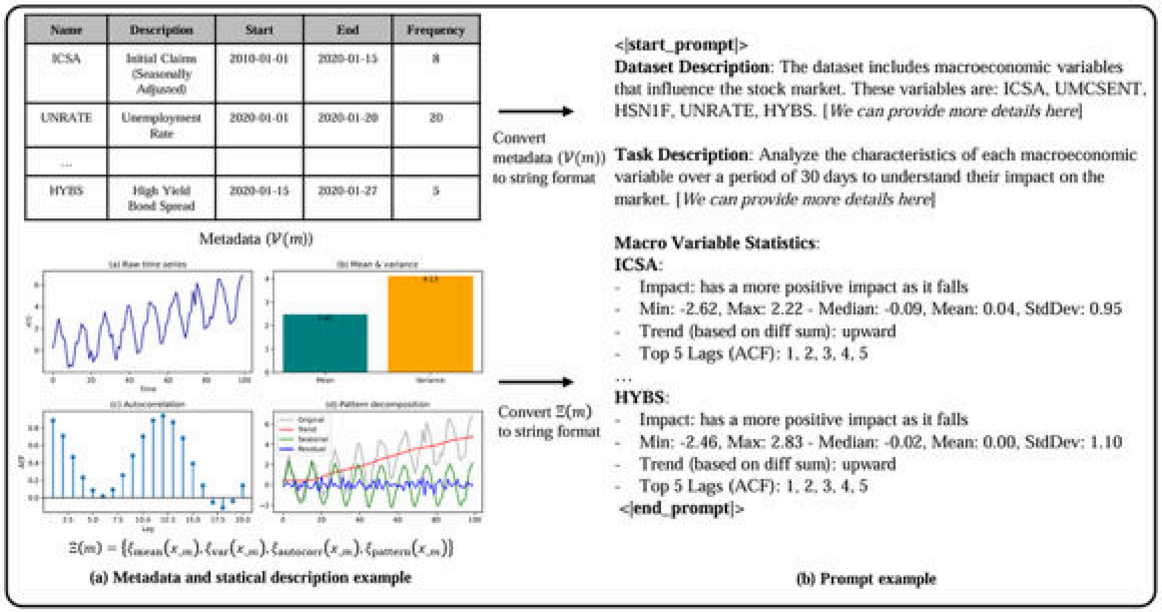

Figure 2.Illustration of LLM-based prompt generation from pairwise outperformance statistics and macroeconomic summaries. For each asset pair, relative performance and sector-level yields are synthesized into textual prompts that capture inter-asset relationships, while similarly constructed macro-level prompts summarize irregularly observed economic indicators. These textual prompts are embedded by the LLM and subsequently integrated, via the cross-attention mechanism, into the DINN architecture.

Hwang, Y., Kong, Y., Zohren, S., & Lee, Y. (2025). Decision-informed neural networks with larg e language model integration for portfolio optimization. arXiv preprint arXiv:2502.00828. https://arxiv.org/abs/2502.00828

Paper 4

August 9–13, 2026
KDD2026
Jeju, Korea

- Text – Time-Series pair dataset

- Pair text and time-series based on semantic and multi-level analysis

FinTexTS: Financial Text-Paired Time-Series Dataset via Semantic-Based and Multi-Level Pairing

Jaehoon Lee[1*], Suhwan Park[1*], Tae Yoon Lim[1], Seunghan Lee[1], Jun Seo[1], Dongwan Kang[1], Hwanil Choi[1], Minjae Kim[1], Sungdong Yoo[1], SoonYoung Lee[1], Yongjae Lee[2], and Wonbin Ahn[1]
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Abstract

The financial domain involves a variety of important time-series problems. Recently, time-series analysis methods that jointly leverage textual and numerical information have gained increasing attention. Accordingly, numerous efforts have been made to construct text-paired time-series datasets in the financial domain. However, financial markets are characterized by complex interdependencies, in which a company's stock price is influenced not only by company-specific events but also by events in other companies and broader macroeconomic factors. Existing approaches that pair text with financial time-series data based on simple keyword matching often fail to capture such complex relationships. To address this limitation, we propose a semantic-based and multi-level pairing framework. Specifically, we extract company-specific context for the target company from SEC filings and apply an embedding-based matching mechanism to retrieve semantically relevant news articles based on this context. Furthermore, we classify news articles into four levels (macro-level, sector-level, related company-level, and target-company level) using large language models (LLMs), enabling multi-level pairing of news articles with the target company. Applying this framework to publicly-available news datasets, we construct **FinTexTS**, a new large-scale text-paired stock price dataset. Experimental results on **FinTexTS** demonstrate the effectiveness of our semantic-based and multi-level pairing strategy in stock price forecasting. In addition to publicly-available news underlying **FinTexTS**, we show that applying our method to proprietary yet carefully curated news sources leads to higher-quality paired data and improved stock price forecasting performance.

CCS Concepts

• Computing methodologies → Language resources; Artificial intelligence; • Applied computing → Economics; Forecasting.

Keywords

Stock Forecasting, Financial Multimodal Dataset, Time Series

*Both authors contributed equally to this research.

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Jaehoon Lee[1*], Suhwan Park[1*], Tae Yoon Lim[1], Seunghan Lee[1], Jun Seo[1], Dongwan Kang[1], Hwanil Choi[1], Minjae Kim[1], Sungdong Yoo[1], SoonYoung Lee[1], Yongjae Lee[2], and Wonbin Ahn[1]. 2026. FinTexTS: Financial Text-Paired Time-Series Dataset via Semantic-Based and Multi-Level Pairing. In . ACM, New York, NY, USA, 14 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

In the financial domain, effectively modeling time-series data has long been considered critically important, as many economic signals, including stock prices, exchange rates, commodity prices, and market volatility, evolve over time. Motivated by this importance, numerous deep learning-based time-series models have been proposed to address financial time-series problems [15, 16, 35]. Among various time-series tasks, stock price forecasting has received particularly strong research attention [9, 29].

Recently, the great success of vision–language multimodal models [8, 10, 19] has sparked growing interest in multimodal learning across a wide range of domains. Accordingly, the development of text-time series (text-TS) multimodal models has also attracted increasing attention. By jointly leveraging numerical time-series data and textual information such as news articles, these text-TS models offer greater potential than traditional approaches solely relying on numerical time-series data [11, 21].

The importance of time-series problems in finance, together with the growing interest in multimodal learning within the time-series community, has led to increased research efforts on constructing multimodal financial datasets as well as developing multimodal learning methods. However, the inherently complex nature of the financial domain makes the construction of text-paired time-series datasets particularly challenging. Specifically, in large-scale financial markets, individual components are interconnected to varying degrees, and these components often form clusters whose collective dynamics influence individual behaviors. For example, a company's stock price is affected not only by events within the company itself, but also by developments involving related companies such as partners or competitors, as well as by broader factors including sector-wide trends and national macroeconomic conditions.

Several prior studies have attempted to construct text-paired time-series datasets specifically for stock price data in the financial domain [26, 27]. However, most existing approaches rely on simple keyword-based matching to pair individual companies' stock prices with related texts, which leads to several limitations. First, such methods fail to capture texts that are semantically relevant to a company when the company is not explicitly mentioned. For instance, news articles discussing the construction of GPU data

Lee, J., Park, S., Lim, T. Y., Lee, S., Seo, J., Kang, D., ... & Ahn, W. (2026). FinTexTS: Financial Text-Paired Time-Series Dataset via Semantic-Based and Multi-Level Pairing. *KDD 2026 Datasets & Benchmarks Track*
https://arxiv.org/abs/2603.02702

Paper 4

FinTexTS: Financial Text-Paired Time-Series Dataset via Semantic-Based and Multi-Level Pairing

Jaehoon Lee[1*], Suhwan Park[1*], Tae Yoon Lim[1], Seunghan Lee[1], Jun Seo[1], Dongwan Kang[1], Hwanil Choi[1], Minjae Kim[1], Sungdong Yoo[1], SoonYoung Lee[1], Yongjae Lee[2], and Wonbin Ahn[1]
LG AI Research[1] & Ulsan National Institute of Science and Technology (UNIST)[2]
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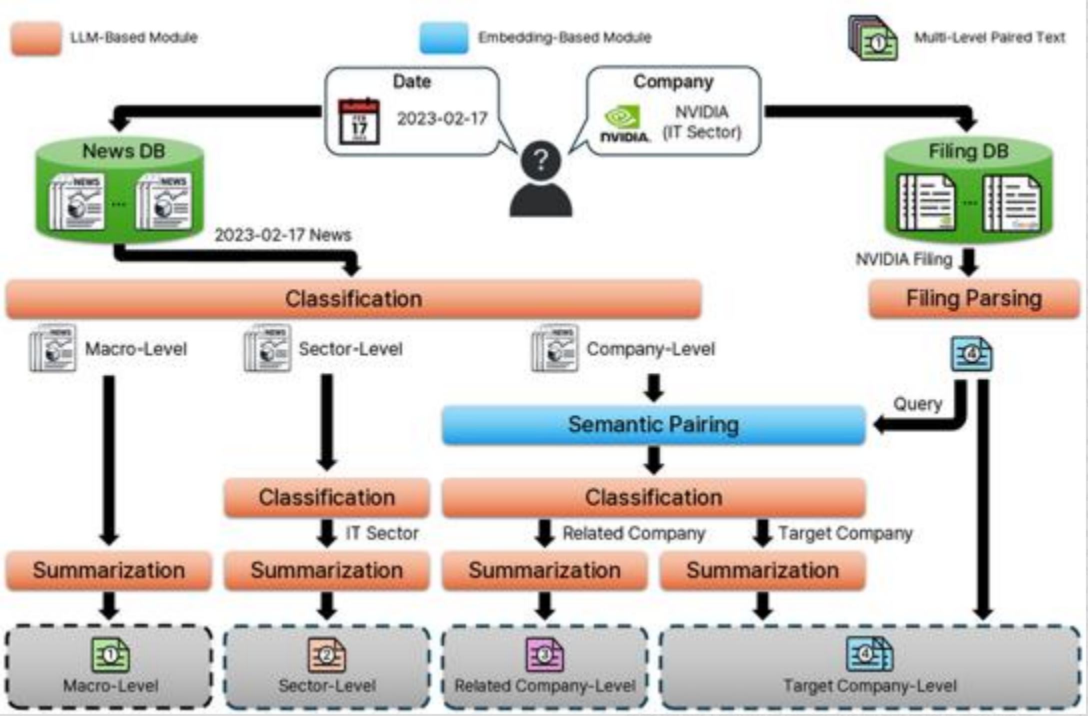
- Pair text and semantic and



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Prediction markets

Prediction markets

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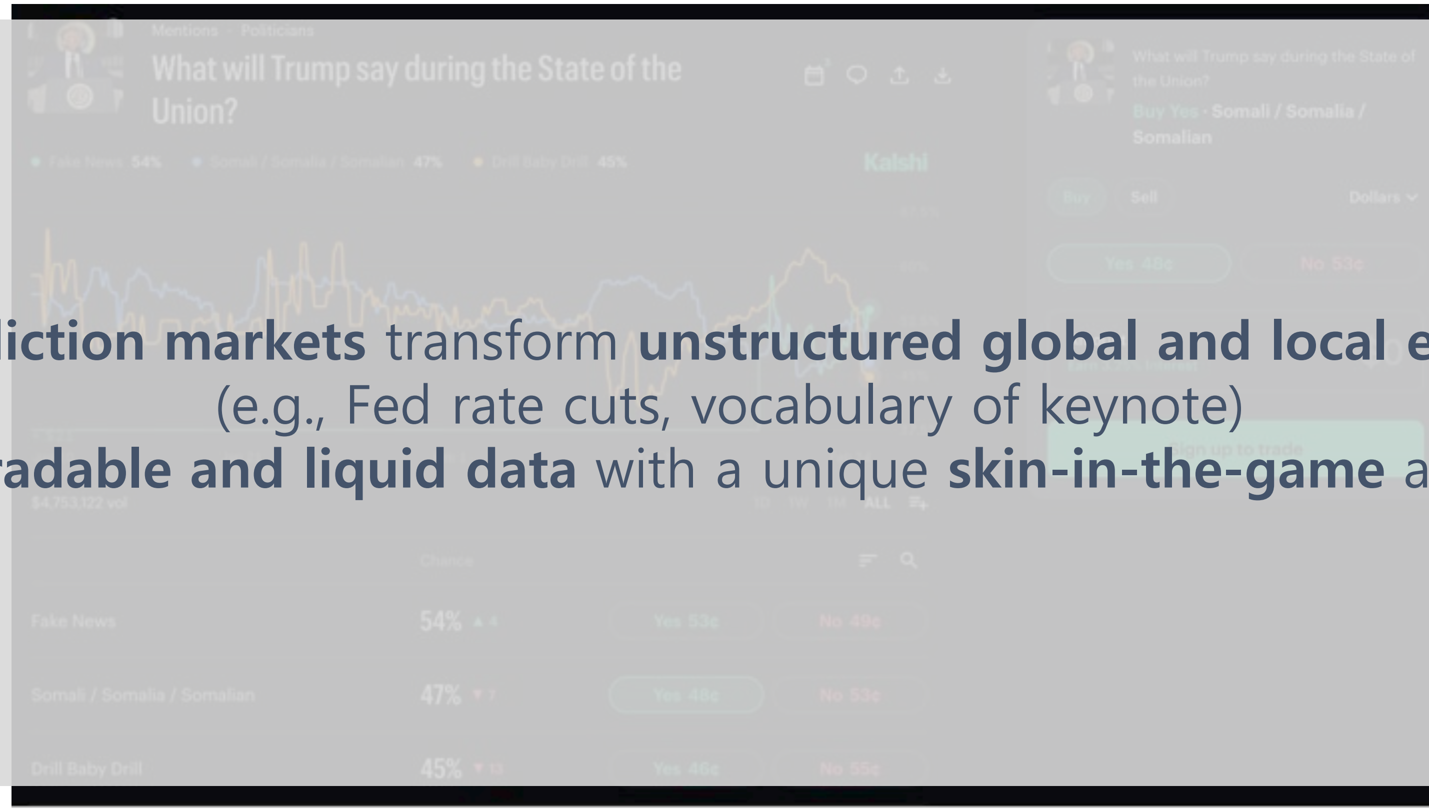

Prediction markets transform **unstructured global and local events** (e.g., Fed rate cuts, vocabulary of keynote) into **tradable and liquid data** with a unique **skin-in-the-game** accuracy

Paper 5

ACL 2026 SAN DIEGO JULY 2-7

- Used LLM to filter and re-rank lead-lag relationships found by Granger causality to improve generalizability
- On Kalshi Economics markets, our hybrid approach consistently outperforms the statistical baseline

LLM AS A RISK MANAGER: LLM SEMANTIC FILTERING FOR LEAD–LAG TRADING IN PREDICTION MARKETS

Sumin Kim[1,*], Minjae Kim[1,*], Jihoon Kwon[1,*], Yoon Kim[2], Nicole Kagan[3], Joo Won Lee[4], Oscar Levy[5], Alejandro Lopez-Lira[6], Yongjae Lee[7,**], Chanyeol Choi[1,**]

[1]LinqAlpha [2]Massachusetts Institute of Technology [3]Kalshi
[4]Arrowpoint Investment Partners [5]University of California, Berkeley
[6]University of Florida [7]UNIST

[*]Equal Contribution. [**]Corresponding authors.

ABSTRACT

Prediction markets provide a unique setting where event-level time series are directly tied to natural-language descriptions, yet discovering robust lead–lag relationships remains challenging due to spurious statistical correlations. We propose a hybrid two-stage causal screener to address this challenge: (i) a statistical stage that uses Granger causality to identify candidate leader–follower pairs from market-implied probability time series, and (ii) an LLM-based semantic stage that re-ranks these candidates by assessing whether the proposed direction admits a plausible economic transmission mechanism based on event descriptions. Because causal ground truth is unobserved, we evaluate the ranked pairs using a fixed, signal-triggered trading protocol that maps relationship quality into realized profit and loss (PnL). On Kalshi Economics markets, our hybrid approach consistently outperforms the statistical baseline. Across rolling evaluations, the win rate increases from 51.4% to 54.5%. Crucially, the average magnitude of losing trades decreases substantially from $649 to $347. This reduction is driven by the LLM's ability to filter out statistically fragile links that are prone to large losses, rather than relying on rare gains. These improvements remain stable across different trading configurations, indicating that the gains are not driven by specific parameter choices. Overall, the results suggest that LLMs function as *semantic risk managers* on top of statistical discovery, prioritizing lead–lag relationships that generalize under changing market conditions.

1 INTRODUCTION

Real-world events rarely occur in isolation; information about one event often changes expectations about what happens next. For example, an inflation surprise can raise expectations of tighter monetary policy. These dependencies often manifest as *lead–lag* structure, where movement in one event systematically precedes movement in another (Bennett et al., 2022).

Identifying such structure from observational time series remains challenging in practice. Traditional statistical methods such as Granger causality can detect directional predictability, but the resulting relationships are often unstable over time and may not reflect meaningful underlying mechanisms (Rossi & Wang, 2019; Phillips, 1986). In large-scale settings, statistically significant lead–lag links

Kim, S., Kim, M., Kwon, J., Kim, Y., Kagan, N., Lee, J. W., Levy, O., Lopez-Lira, A., Lee, Y. & Choi, C. (2026). LLM as a Risk Manager: LLM Semantic Filtering for Lead-Lag Trading in Prediction Markets. *The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026), Industry Track, Accepted*
https://arxiv.org/abs/2602.07048

Paper 5

LLM AS A RISK MANAGER: LLM SEMANTIC FILTERING FOR LEAD–LAG TRADING IN PREDICTION MARKETS

Sumin Kim[1,*], Minjae Kim[1,*], Jihoon Kwon[1,*], Yoon Kim[2], Nicole Kagan[3], Joo Won Lee[4], Oscar Levy[5], Alejandro Lopez-Lira[6], Yongjae Lee[7,**], Chanyeol Choi[1,**]

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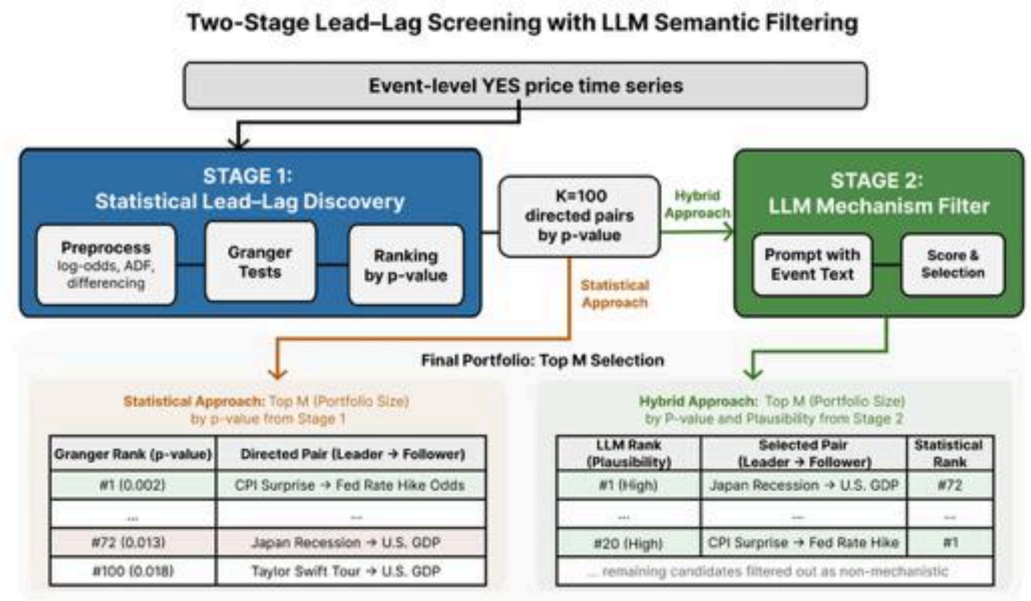

Figure 1: Two-stage framework for leader–follower pair discovery in prediction markets. Stage 1 produces a candidate set of Top K directed pairs (K=100) ranked by Granger significance, and Stage 2 applies LLM-based semantic re-ranking to select the final Top M portfolio (M=20).

..., O., Lopez-Lira, A., Lee, Y. & Choi, C. (2026). LLM as a Risk Manager: LLM Semantic Filtering for Lead-Lag Trading in Prediction Markets. *The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026), Industry Track, Accepted*
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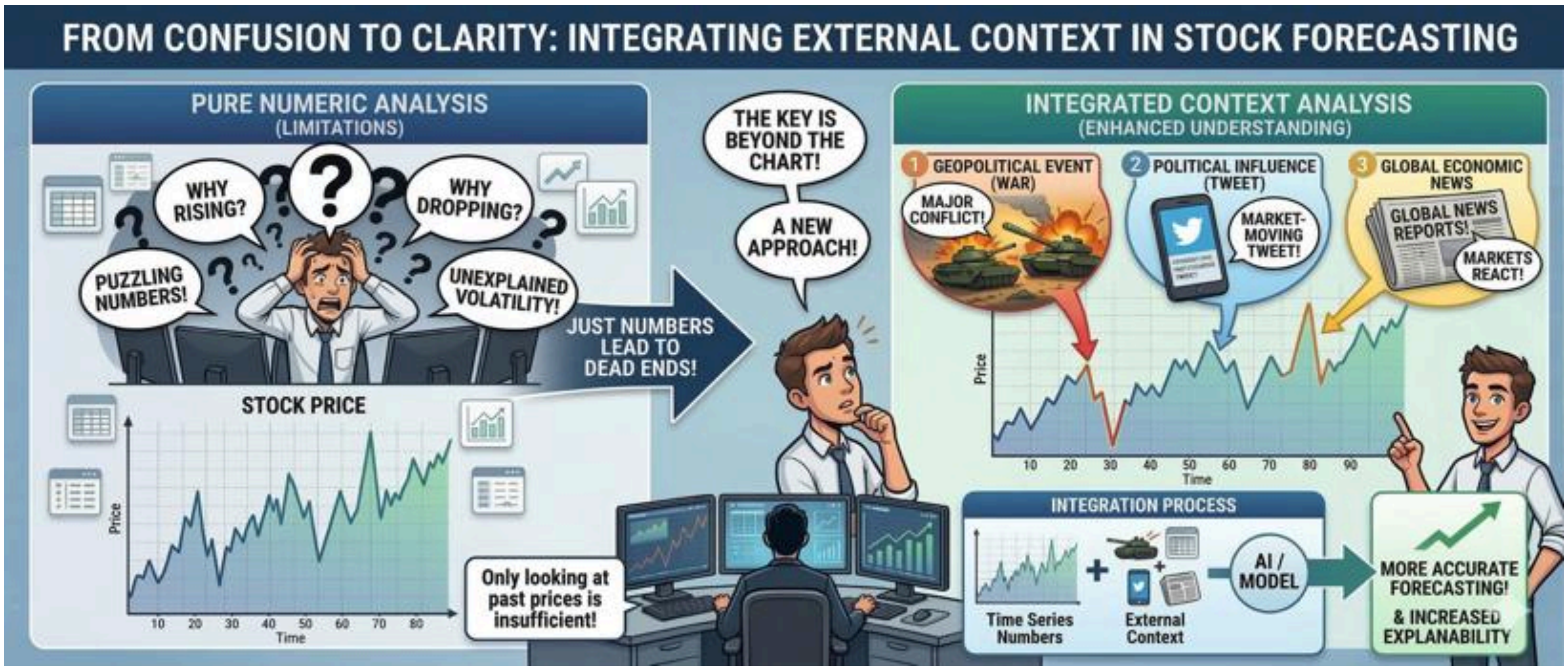
FROM CONFUSION TO CLARITY: INTEGRATING EXTERNAL CONTEXT IN STOCK FORECASTING
PURE NUMERIC ANALYSIS
(LIMITATIONS)
WHY RISING?
WHY DROPPING?
PUZZLING NUMBERS!
UNEXPLAINED VOLATILITY!
STOCK PRICE
Price
Time
JUST NUMBERS LEAD TO DEAD ENDS!
Only looking at past prices is insufficient!
THE KEY IS BEYOND THE CHART!
A NEW APPROACH!
INTEGRATED CONTEXT ANALYSIS
(ENHANCED UNDERSTANDING)
1 GEOPOLITICAL EVENT (WAR)
MAJOR CONFLICT!
2 POLITICAL INFLUENCE (TWEET)
MARKET-MOVING TWEET!
3 GLOBAL ECONOMIC NEWS
GLOBAL NEWS REPORTS!
MARKETS REACT!
INTEGRATION PROCESS
Time Series Numbers
External Context
AI / MODEL
MORE ACCURATE FORECASTING!
& INCREASED EXPLANABILITY